Web Appendix for
Levels and trends in fertility rates among adolescents aged 15–19 in 21 States and Union Territories in India from 1990 to 2050: A Bayesian Modelling Study

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May 28, 2026

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Please cite this web appendix using its DOI:10.6084/m9.figshare.30772265

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List of Abbreviations

AR1	first-order Autoregressive
ASFR	Age-specific Fertility Rate
CPO	Conditional Predictive Ordinates
DHS	Demographic and Health Survey
DIC	Deviance Information Criterion
GATHER	Guidelines for Accurate and Transparent Health Estimates Reporting
ICAR	Intrinsic Conditional Autoregressive
INLA	Integrated Nested Laplace Approximation
JAGS	Just Another Gibbs Sampler
MCMC	Markov chain Monte Carlo
NFHS	National Family Health Survey
PC	Penalized Complex
PI	Prediction Interval
SRS	Sample Registration System
TFR	Total Fertility Rate
UN	United Nations
UT	Union Territory
WAIC	Watanabe-Akaike Information Criterion
WPP	World Population Prospects

Figure 1 presents an overview of the database construction and Bayesian model process for this study.

1 Data

We compiled the database of age-specific fertility rate (ASFR) for adolescent females aged 15–19 across the 29 largest Indian states/UTs.[1] Table 1 lists the 29 Indian states/UTs included in our study. Among them, 21 are classified as larger states/UTs (with population more than 10 million as per Census 2011) for which we present the state-level results.

	[21] Larger states/UTs	[8] Smaller states/UTs
Indian States/UT names	former state of Andhra Pradesh (including Telangana); Assam; Bihar; Chhattisgarh; Delhi; Gujarat; Haryana; Himachal Pradesh; Jammu and Kashmir (including Ladakh); Jharkhand; Karnataka; Kerala; Madhya Pradesh; Maharashtra; Odisha; Punjab; Rajasthan; Tamil Nadu; Uttar Pradesh; Uttarakhand; West Bengal	Arunachal Pradesh; Goa; Manipur; Meghalaya; Mizoram; Nagaland; Sikkim; Tripura

Table 1: **Indian states/UTs classification by population size.** The red numbers at the beginning of a cell refer to the number of states/UTs that fall under that category. Bigger states/UTs: with population more than 10 million as per Census 2011.

Table 2 provides an overview of the Indian state-level ASFR 15–19 observations by different data series. There are 1,510 observations available from all states/UTs in India. The reference years of the observations range from 1967 to 2020. The database is publicly available at figshare repository [2]

Data Type	Series Name (Series Year)	# Obs.	# states/UTs
DHS	NFHS 1992–1993	207	29
	NFHS 1998–1999	216	
	NFHS 2005–2006	252	
	NFHS 2015–2016	261	
	NFHS 2019–2021	261	
SRS	–	313	26
Total		**1,510**	**29**

Table 2: **Observation distribution by source type for ASFR 15–19.** DHS: Demographic and Health Surveys. NFHS: National Family Health Survey, refers to India Demographic and Health Surveys. SRS: Sample Registration System.

1.1 Age-specific fertility rate 15–19 data pre-processing

Data from full birth histories are extracted from the five National Family Health Survey (NFHS; i.e. India Demographic and Health Surveys) 1992–1993, 1998–1999, 2005–2006, 2015–2016, and 2019–2021. We compute the observations (Section 1.1.1) and corresponding sampling errors (Section 1.1.2) of ASFR for mothers aged 15–19

[1] Telangana is combined with Andhra Pradesh because it separated from Andhra Pradesh only in 2014. Hence, we use the name "former state of Andhra Pradesh" to refer to the combination of Andhra Pradesh and Telangana in our study. Jammu and Kashmir includes Ladakh (bifurcated in 2019)

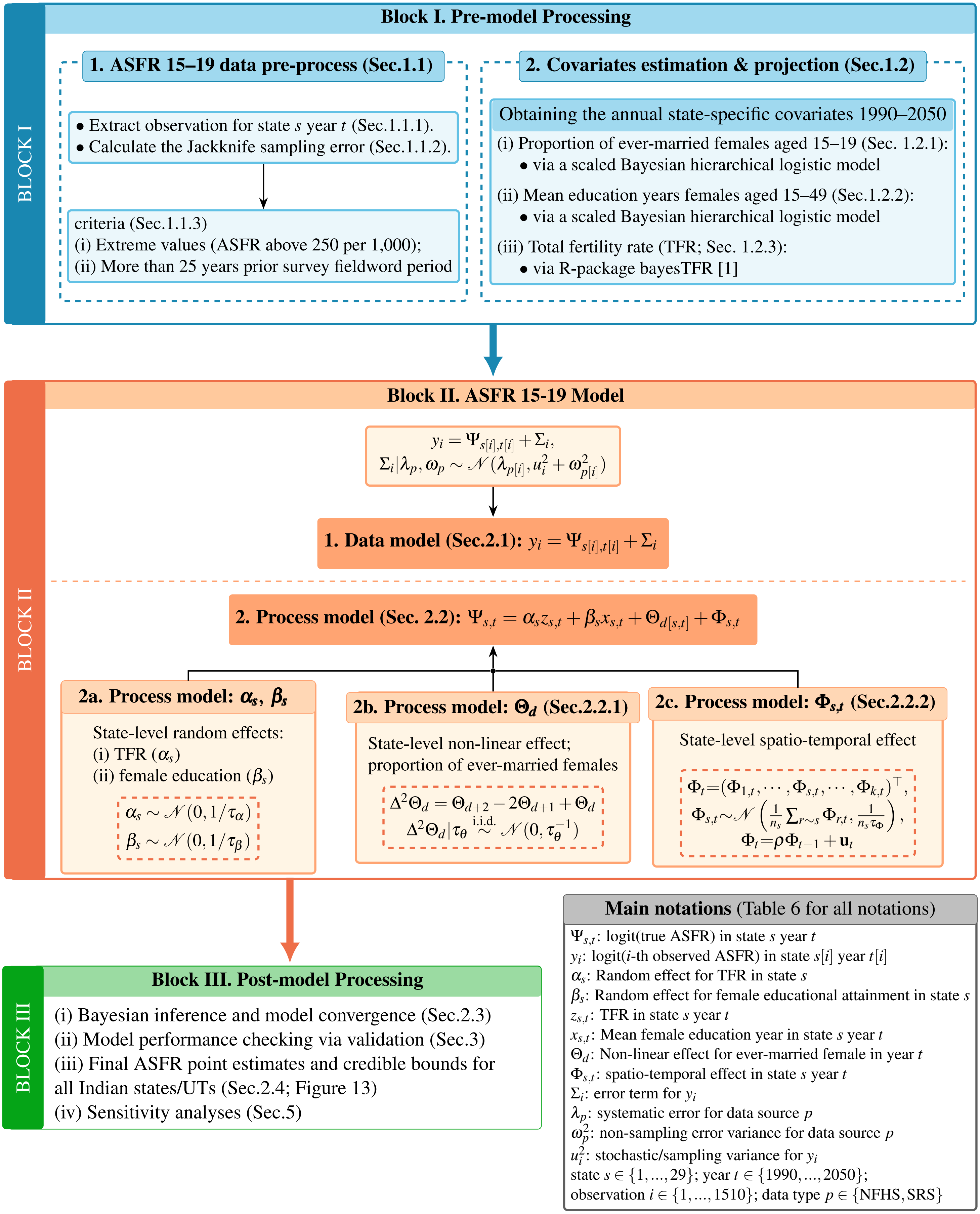

Figure 1: **Flowchart for data and model process overview.** This flow chart summarizes the main steps of pre-model processing (Block I), Bayesian modelling (Block II), where $\Psi_{s,t}$ refers to the true age-specific fertility rate 15–19 on the logit-scale, and the post-model processing (Block III).

from these NFHS. The database for model fitting (summarized in Section 1; DOI:10.6084/m9.figshare.30771611) is based on multiple data quality assessments and pre-processing steps as detailed in the rest of this section.

1.1.1 Extracting observations by data source type

Birth and sibling histories in surveys NFHS data were used to compute fertility rates for adolescent females aged 15–19, for a fixed period of three years for all fertility observations. For simplicity, all notations in this subsection refer to women from age group 15–19, and hence the age group index is omitted.

Let $B_{s,t}$ be the observed number of total births born by mothers within the age group [15, 19] at the time of birth for a certain Indian state/UT s in year t. $N_{s,t}^{\text{female}}$ refers to the number of woman-year exposure in the age group [15, 19]. The observed ASFR for a certain Indian state/UT s year t, $z_{s,t}$ is computed as:

$$z_{s,t} = B_{s,t}/N_{s,t}^{\text{female}}.$$

Birth records from Sample Registration System Data are available in the majority of Indian states/UTs from the sample registration system (SRS). Among the 29 states/UTs we modeled, Jharkhand, Chhattisgarh, and Uttarakhand do not have SRS data.

Given that the SRS data are considered as administrative records and with a high-level of completeness across states over time, we assume that the coefficient of variation of all SRS observations is at 0.01, expressed as $\text{CV}_{\text{ST}} = 0.01$, where the subscript $*_{\text{ST}}$ refers to "'stochastic'. Consequently, the associated stochastic errors of SRS data are imputed as $\sigma_{\text{ST}} = z \cdot \text{CV}_{\text{ST}}$, where z denotes any observation from SRS.

1.1.2 Sampling variance calculation

NFHS adopted a two-stage stratified sampling frame for rural areas and a three-stage sample design for urban areas, except for the 2015–2016 and 2019–2021 NFHS, where a two-stage sampling frame was applied to both rural and urban areas [3, 4]. The first stage is to sample the primary sampling units with probability proportional to population size, followed by random selection of census enumeration blocks and/or households. To simplify, all notations in this section refer to the state level in India, for a specific NFHS. Hence, we remove the subscript s from all notations in this section.

The ASFR 15–19 extraction approach from NFHS data is mainly based on the approach described in [5]. For a specific NFHS, we calculate the Jackknife sampling error for ASFR 15–19 for each consecutive three-year window period prior to the time of interview (i.e. from three years prior interview to date of interview, from six to three years prior interview, etc.). The reference year t of the ASFR 15–19 for an NFHS is taken as the midpoint of the period. The u-th partial prediction of ASFR 15–19 observation from a certain state-specific survey within a certain reference three-year period is given by:

$$z_{-u} = \frac{\sum_{n=1}^{N} \mathbb{I}(15 \leq a_n \in [t-1.5, t+1.5) \leq 19; d_n \neq u) \cdot w_n}{\sum_{m=1}^{M} N_m \mathbb{I}(15 \leq a_m \in [t-1.5, t+1.5) \leq 19; d_n \neq u) \cdot w_m \cdot q_m}, \text{ for } u \in \{1, \ldots, U\},$$

where n indexes the live births in the survey-year, N is the total number of live births for a particular survey-year. $a_n \in [t-1.5, t+1.5)$ refers to the age of month at time of the birth that happened during the three-year period of $[t-1.5, t+1.5)$ with the mid-point of the period t as the reference year, d_n refers to the cluster number with a total of U clusters, and w_n the sampling weight for the n-th live birth. In the denominator, the index m refers to each woman (regardless of giving birth or not) in the sample. N_m denotes the woman-year exposure during the period $[t-1.5, t+1.5)$. The exposures are only summed for women at age $15 \leq a_m \leq 19$. w_m is the sampling weight for the m-th woman. q_m is the total-woman adjustment factor applied to ever-married woman samples (i.e. two surveys NFHS 1992–1993 and 1998–1999) for the m-th woman so that the sampled ever-married woman exposure can be approximated as the exposure for all the women. For NFHS phases III to V, we have $w_m = 1$. The indicator $\mathbb{I}(\cdot) = 1$ if the condition inside the brackets is true, otherwise $\mathbb{I}(\cdot) = 0$. The u-th pseudo-value estimate of ASFR 15–19 in a particular survey-year is:

$$
\begin{aligned}
z_u^* &= U \cdot z - (U-1) \cdot z_{-u}, \text{ where} \\
z &= \frac{\sum_{n=1}^{N} \mathbb{I}(15 \leq a_n \in [t-1.5, t+1.5) \leq 19) \cdot w_n}{\sum_{m=1}^{M} N_m \mathbb{I}(15 \leq a_m \in [t-1.5, t+1.5) \leq 19) \cdot w_m \cdot q_m}.
\end{aligned}
$$

The Jackknife standard error is:

$$
\sigma_{\text{JK}} = \sqrt{\frac{\sum_{u=1}^{U} \left(z_u^* - \overline{z^*}\right)^2}{U(U-1)}}, \text{ where } \overline{z^*} = \frac{1}{U} \sum_{u=1}^{U} z_u^*.
$$

1.1.3 Exclusion criteria

Before fitting the ASFR 15–19 observations to the Bayesian model, the state analysts reviewed the data sources and data points available. Some of the data points were excluded due to concerns about data quality and completeness. We used the following exclusion criteria to ensure data quality used in our model:

- Exclude ASFR 15–19 data with outlying high values. Specifically, we only keep observations below 250 per 1,000;
  - The upper cutoff value is based on the United Nation database of ASFR 15–19 from all countries over time [6]. 250 births per 1,000 mothers aged 15–19 marks the top 0.05th percentile. The ASFR 15–19 observations above this value are usually subject to low data quality.
  - Eight data points were excluded under this criterion, corresponding to 0.5% of the total number of data points.
- For survey data, exclude data points beyond 25 years prior to the survey fieldwork period.
  - All data points are within 25 years prior survey time. Hence, zero data were excluded under this criterion.

1.2 Covariates imputation

When estimating and projecting ASFR 15–19 across Indian states/UTs up till 2050, we incorporated three external variables in our state-level ASFR process model (Section 2.2). All covariates are Indian state-year-specific:

1. Section 1.2.1: Proportion of ever-married females aged 15–19;
2. Section 1.2.2: Mean education years among females aged 15–49; and
3. Section 1.2.3: Total fertility rate.

Figure 2 illustrates the exploratory analyses of the potential relationships between the aforementioned variables against the outcome of interest state-level ASFR 15–19. The exploratory loess curves in Figure 2 are not equivalent to reflect the actual effects in the model. This is due to the fact that each plot solely illustrates the relationship between ASFR 15–19 and one covariate, not conditioning on the other two variables. Furthermore, the plots present overall patterns across all districts. We chose Indian districts instead of Indian state/UT as the measurement unit in this plot so that we have more data points to motivate the variable choices. However, the measurement unit differs from the unit we model the outcome of interest: state-specific patterns. Hence, the primary purpose of this plot is to (i) motivate the variable choice (with non-horizontal loess curves), and (ii) how strong the relationship is between the variables: the adjusted-like R^2 values ($R^2_{\text{adj-like}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2/(n-df-1)}{\sum_i (y_i - \bar{y})^2/(n-1)}$, with y_i the logit-scaled ASFR, $\hat{y}_i$ the LOESS fitted value of y_i, $\bar{y} = \sum_i y_i/n$, n the number of observations, df the effective number of parameters per LOESS fit) closer to one indicate stronger relationship. Figure 2 shows that the relation between ASFR 15–19 and the proportion of married females age 15–19 (with $R^2_{\text{adj-like}} = 0.75$) is much stronger than the other two variables ($R^2_{\text{adj-like}}$ are 0.39 and 0.16). It may not, however, be used to imply the type of relationship used in the ASFR model.

Before we get into the modeling of the ultimate outcome of interest which is state-level ASFR for females aged 15–19 in India, we need the three variables for each state/UT annually from 1990 to 2050. The rest of this section explains how we imputed the values of these variables given that none of the variables is available for all state-years.

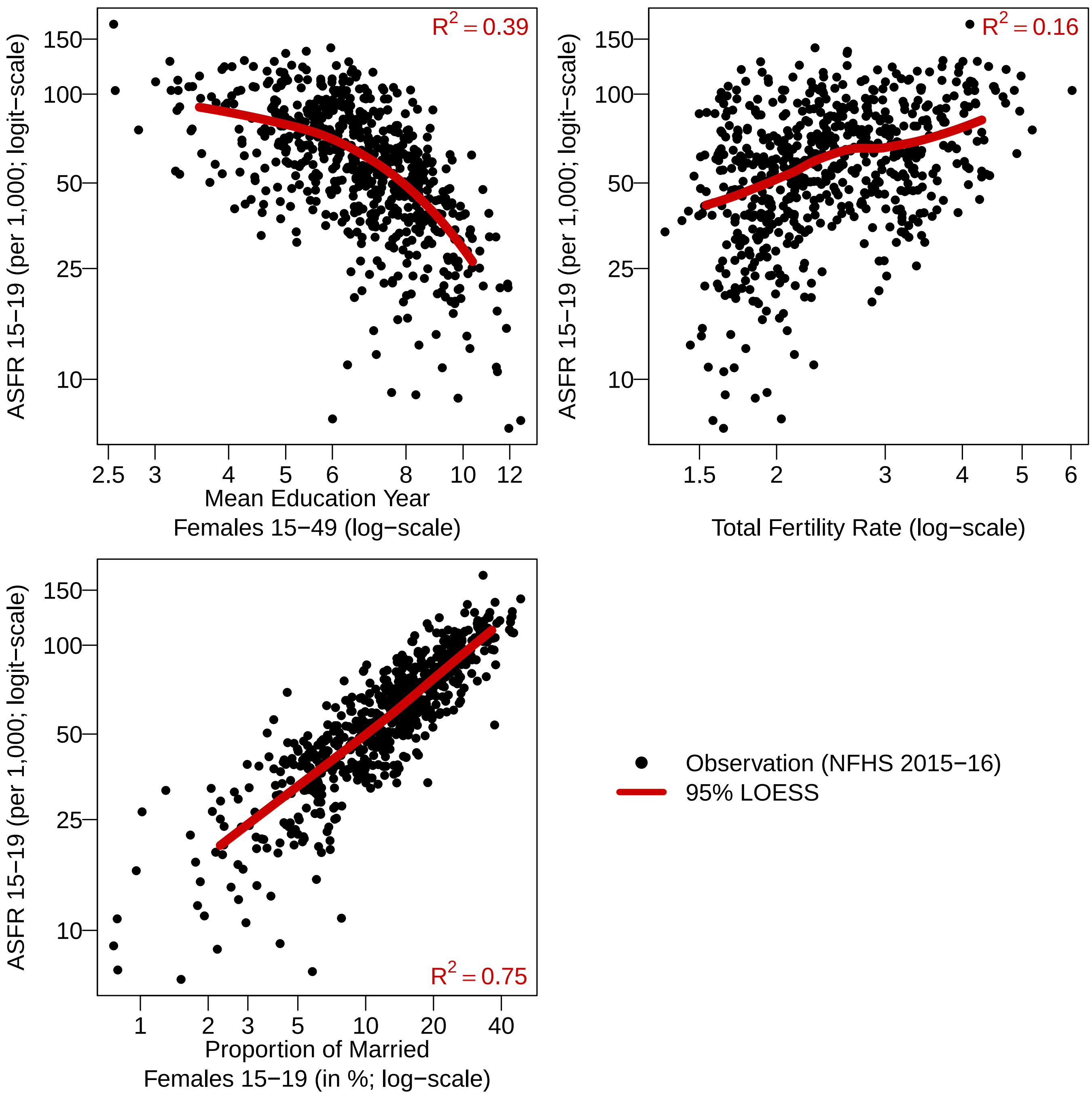

Figure 2: **Motivation of variable choice in model.** Red curve: 95% LOESS fit based on observations between 2.5th and 97.5th percentiles of each variable. Black dots: 640 district-level observations from the NFHS 2015–16. LOESS: Local Polynomial Regression Fitting. Adjusted-like R^2 for each LOESS fit is shown as red text in each plot. X-axis and y-axis are on log and logit scales respectively to align with the scales used in the model.

1.2.1 Proportion of ever-married female aged 15–19 across Indian states/UTs overtime

The proportion of ever-married females aged 15–19 in Indian state/UT s in year t is denoted as $V_{s,t}$, which is used as an input variable later on in Section 2.2. The resulting $V_{s,t}$ are illustrated in Figure 10. We obtained its annual estimates and projections through a scaled Bayesian hierarchical logistic model. Specifically, for the j-th observed proportion of ever-married female aged 15–19 v_j from Indian state $s[j]$ in year $t[j]$, we assume it follows a normal distribution on the log-scale:

$$v_j \sim \mathcal{N}(V_{s[j],t[j]}, \omega_v^2), \text{ for } j \in \{1, \cdots, 125\}.$$

v_j denotes the j-th reported proportion of ever-married females aged 15–19 from the five NFHS. We applied the "all woman factor" to NFHS 1992–93 and 1998–99 surveys since they only collected information from ever-married females. We processed NFHS microdata and compiled a database with 133 observations of proportion of ever-married females aged 15–19. The detailed data sources for each Indian state/UT is in Table 13.

The variance of the distribution is ω_v^2, the non-sampling variance parameter for all the NFHS observations (hence estimated in the model), representing the data errors that are not possible to quantify or be eliminated mainly due to non-response, recall errors, and data recording errors. We assigned an informative prior to ω_v by assuming that the true underlying state-level proportion of ever-married female aged 15–19 are close to the NFHS observations with little deviation away from them.

The mean of the distribution $V_{s[j],t[j]}$ is the true proportion of ever-married female aged 15–19 on the log-scale for India in state $s[j]$ in year $t[j]$ for the j-th observation. The mean $V_{s,t}$ is modeled as a scaled logistic function with independent variable $\log(t)$ log of time index and state-specific parameters that define the features of a logit curve, including coefficients λ_s (for rate of decline), intercept parameters ζ_s (for the average level), and the scale parameter δ_s which models the maximum proportion of ever-married female aged 15–19 for each state s. We have

$$\exp\{V_{s,t}\} = \frac{\delta_s^{\text{mar}} \cdot \exp\{\lambda_s^{\text{mar}} \cdot \log(t) + \zeta_s^{\text{mar}}\}}{1 + \exp\{\lambda_s^{\text{mar}} \cdot \log(t) + \zeta_s^{\text{mar}}\}}, \text{ for } \forall s, \forall t.$$

All state-specific parameters were assigned with hierarchical distributions so that the logit curve patterns can be pooled towards an average shape when a certain state lack data, and can still follow observations wherever available. Specifically, we used a truncated hierarchical normal distribution for δ_s by assigning an upper bound of $\log(100)$ (i.e. the upper bound of 100% on the original scale) to reflect the fact that early marriage in India has long been prepubertal historically [7]. The shape-determinant parameters are modeled as

$$\begin{aligned}
\delta_s^{\text{mar}} &\sim \mathcal{N}_{(-\infty, \log(100))}\left(\mu_\delta^{\text{mar}}, \left(\sigma_\delta^{\text{mar}}\right)^2\right), \text{ for } \forall s, \\
\lambda_s^{\text{mar}} &\sim \mathcal{N}\left(\mu_\lambda^{\text{mar}}, \left(\sigma_\lambda^{\text{mar}}\right)^2\right), \text{ for } \forall s, \\
\zeta_s^{\text{mar}} &\sim \mathcal{N}\left(\mu_\zeta^{\text{mar}}, \left(\sigma_\zeta^{\text{mar}}\right)^2\right), \text{ for } \forall s.
\end{aligned}$$

We assigned non-informative priors to the global parameters μ_δ^{mar}, μ_λ^{mar}, μ_ζ^{mar}, $\sigma_\delta^{\text{mar}}$, $\sigma_\lambda^{\text{mar}}$, and $\sigma_\zeta^{\text{mar}}$.

1.2.2 Mean education years in females aged 15–49 across Indian states/UTs overtime

The mean education years among females aged 15–49 in Indian state/UT s in year t, denoted as $x_{s,t}$ later on in Section 2.2, are model results from a Bayesian hierarchical model. The resulting $x_{s,t}$ are illustrated in Figure 11.

The model inputs were the reported mean education years among females aged 15–49 from the five NFHS. We applied the "all woman factor" to NFHS 1992–93 and 1998–99 surveys since they only collected information from ever-married females. We processed NFHS microdata files and compiled a database with 132 observations of mean education years among females aged 15–49. The detailed data sources for each Indian state/UT is in Table 14.

We obtained the annual estimates and projection through a Bayesian hierarchical scaled-logistic model. Specifically, for the i-th observed log of female mean education years x_i from Indian state $s[i]$ in year $t[i]$, we assume it follows a normal distribution on the log-scale:

$$
\begin{aligned}
x_i &\sim \mathcal{N}(\chi_{s[i],t[i]}, \omega_\chi^2), \text{ for } i \in \{1, \cdots, 132\},\\
\exp\{\chi_{s,t}\} &= \frac{\delta_s^{\text{edu}} \cdot \exp\left\{\lambda_s^{\text{edu}} \cdot \log(t) + \zeta_s^{\text{edu}}\right\}}{1 + \exp\{\lambda_s^{\text{edu}} \cdot \log(t) + \zeta_s^{\text{edu}}\}}, \text{ for } \forall s, \forall t,\\
\delta_s^{\text{edu}} &\sim \mathcal{N}_{(-\infty, \log(20))}\left(\mu_\delta^{\text{edu}}, \left(\sigma_\delta^{\text{edu}}\right)^2\right), \text{ for } \forall s,\\
\lambda_s^{\text{edu}} &\sim \mathcal{N}\left(\mu_\lambda^{\text{edu}}, \left(\sigma_\lambda^{\text{edu}}\right)^2\right), \text{ for } \forall s,\\
\zeta_s^{\text{edu}} &\sim \mathcal{N}\left(\mu_\zeta^{\text{edu}}, \left(\sigma_\zeta^{\text{edu}}\right)^2\right), \text{ for } \forall s.
\end{aligned}
$$

The mean of the distribution $\chi_{s[i],t[i]}$ is the true female mean education years on the log-scale for India in state $s[i]$ in year $t[i]$ for the i-th observation. The the mean is modeled as a scaled logistic function with independent variable $\log(t)$ log of time index. All state-specific parameters that define the features of a logit curve were assigned with hierarchical normal distributions with global means and variances. The distribution for the state-level scale parameters is truncated with upper limit of $\log(20)$ by assuming that it is highly unlikely to have more than 20 years of education for females aged 15–49. The variance of the distribution is ω_χ^2 with an informative prior assigned to ω_χ by assuming that the true underlying state-level female mean education years follow closely to the NFHS observed levels and trends.

1.2.3 Total fertility rate across Indian states/UTs overtime

The reported total fertility rate (TFR) in Indian states/UTs was obtained through the bayesTFR [1] subnational Bayesian hierarchical model. Input data for modelling the subnational TFR in India is detailed in [8] and can be directly downloaded from the BayesPop webpage (bayesTFR is one of the subsidiary projects under the BayesPop project). The original input data for modeling were in a five-year interval with the most recent observations in 2010, containing 510 data points. We supplemented the original data file with NFHS survey data up to reference year 2020. The final input database we used to fit the bayesTFR model has 638 observations.

The bayesTFR model generated state-level TFR estimates and projections at a five-year interval, assuming the state-specific TFR is a multiplicative ratio of the national TFR in the same year. To obtain annual results, simulated fertility trajectories were generated through annual interpolation between the five-year knots. From these trajectories, we obtained annual posterior medians and 95% credible intervals for each state and year from 1950 to 2050. The TFR observations and model results are in Figure 12.

2 Methods

In Section 2.1 and Section 2.2, we summarize the model to estimate fertility rate for age group 15–19. The logit-scaled ASFR 15–19 is modeled as the combination of the effect of female marital status, female education attainment, TFR, and temporal correlation.

The outcome of interest is the ASFR for age group 15–19 in Indian state/UT s in year t, where we model it on the logit scale and is denoted as $\Psi_{s,t}$. Throughout the report, $\mathcal{N}(\mu, \sigma^2)$ refers to a normal distribution with mean μ and variance σ^2 and $\mathcal{U}(a,b)$ refers to a continuous uniform distribution with lower and upper bounds at a and b respectively. The notations and indexes are summarized in Table 6. The detailed explanations of the model is in the rest of this section.

2.1 Data model

Let z_i to be the i-th observed ASFR 15–19 and $y_i = \text{logit}(z_i)$. We assumed the following:

$$\begin{aligned} y_i &= \Psi_{s[i],t[i]} + \Sigma_i, \text{ for } i \in \{1, \cdots, n\}, & (1)\\ \Sigma_i | \lambda_p, \omega_p &\sim \mathscr{N}(\lambda_{p[i]}, u_i^2 + \omega_{p[i]}^2), & (2)\\ \lambda_p &\sim \mathscr{N}(0, 1/\tau_\lambda), \text{ for } p \in \{1,2\}, & (3)\\ \omega_p^{-2} &\sim \mathscr{PC}(z, 0.01), \text{ for } p \in \{1,2\}. & (4) \end{aligned}$$

For women aged 15–19, the ASFR observations are indexed by $i \in \{1, \cdots, n\}$. y_i denotes the i-th logit of observed ASFR in state $s[i]$ in year $t[i]$ and is modeled on the logit scale. $1/(1+\exp\{-\Psi_{s,t}\})$ is the outcome of interest, the true ASFR in Indian state/UT s in year t. Σ_i is the error term for y_i. The error term Σ_i follows a normal distribution as shown in Equation 2.

The mean of the error term is assumed to vary by data source type $p \in \{1,2\}$ where $p = 1$ refers to NFHS and $p = 2$ refers to SRS. The source-specific random effects λ_p account for the underlying systematic discrepancies in NFHS and SRS that are unmeasurable in the data pre-processing [9]. λ_p reflects the deviations that are systematic for each data source. Hence, λ is assumed to differ by data source type $p \in \{1,2\}$ and assumed stable and the same among Indian states/UTs. All the state-specific effects are modeled in Section 2.2. Table 3 presents the model results for λ_p. The results are consistent with the observation such that the SRS data are systematically below the NFHS data for most periods. The fact that the data model include the parameter λ_p imply the model estimates for ASFR is pooled in-between the two source types as shown in the state model results in Figure 13. In Figure 3, we compare the density of the residuals and the adjusted residuals by data source type. Figure 3 (top) implies a clear separation between the residual densities from NFHS and SRS. The model estimates are pooled in-between the two source types. Figure 3 (bottom) shows that conditioning on the source-specific error parameter λ_p, the adjusted residuals are centered around zero for both data sources, indicating an effective adjustment via model setup.

$\boldsymbol{\lambda}$	**Posterior median**	**95% CI**
NFHS ($p = 1$)	0.45	[0.27; 1.18]
SRS ($p = 2$)	-0.45	[-1.18; -0.27]

Table 3: **Model results for λ_p.**

The variance is the sum of two parts: (i) known stochastic/sampling error variance u_i^2 for the i-th observation, and (ii) unknown non-sampling error variance $\omega_{p[i]}^2$ for data source type p in which the i-th observation belongs. The sampling variance is pre-calculated using the Jackknife resampling method as explained in Section 1.1.2 to account for the uncertainty due to the sample design of the surveys. The non-sampling error variances reflect the uncertainty due to non-response, errors made in data input, recall biases, etc., which can be minimized but difficult to eliminate. We assume it differ across data source types $p \in \{1,2\}$. z is the standard deviation of y_i. We assign Penalized Complexity (PC) priors for the precision parameter ω_p^{-2} for each data source type $p \in \{1,2\}$ [10]. Given such data model, the Bayesian estimates are pooled towards informative observations and are less influenced by weakly informative observations.

PC priors are default priors assigned to precision parameters used in the `R`-package `INLA` [11], which we used for Bayesian inference (refer to Section 2.3 for information regarding the statistical computation). The main advantages of PC priors are that they are invariant to reparameterizations and have excellent robustness properties according to [10].

2.2 Process model of state-level ASFR 15–19

The process model for $\Psi_{s,t}$, the ASFR for age group 15–19 on logit scale, is defined as:

$$\Psi_{s,t} = \alpha_s z_{s,t} + \beta_s x_{s,t} + \Theta_{d[s,t]} + \Phi_{s,t}, \quad (5)$$

where

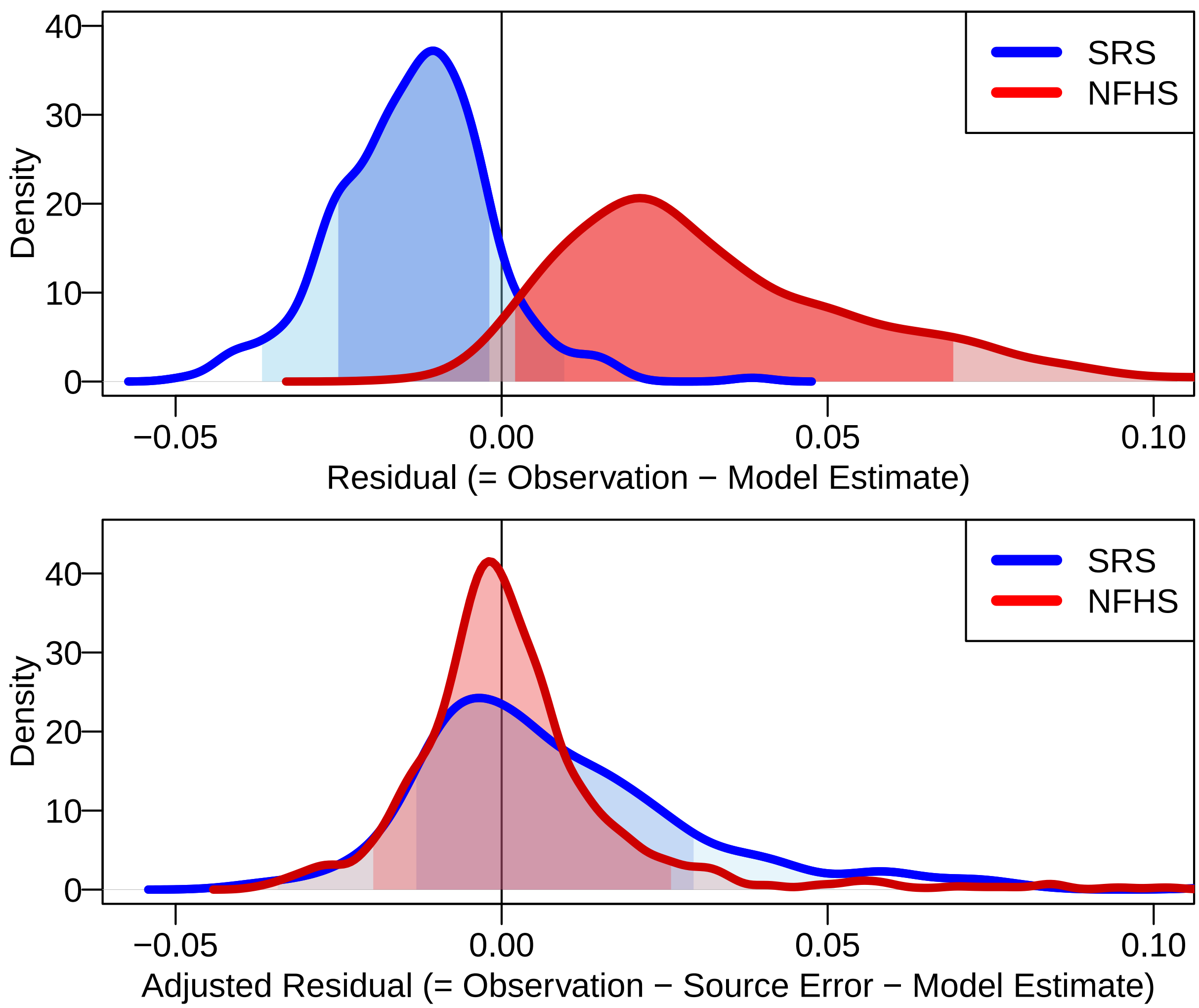

Figure 3: **Residual distribution by data source type.** Top: residual $= \text{logit}^{-1}(y_i) - \text{logit}^{-1}(\Psi_{s[i],t[i]})$, the difference between observation and the model estimates. Bottom: adjusted residual $= \text{logit}^{-1}(y_i - \lambda_{p[i]}) - \text{logit}^{-1}(\Psi_{s[i],t[i]})$, the difference between error-adjusted observation and the model estimates. Blue: residual density for SRS data. Red: residual density for NFHS data. Darker shades: residual $\pm$ SD. Lighter shades: residual $\pm$ 2SD.

- $z_{s,t}$: total fertility rate (TFR) for Indian state/UT s in year t on log scale (refer to Section 1.2.3 for details);
- $x_{s,t}$: mean education years among all females aged 15–49 for Indian state/UT s in year t on log scale (refer to Section 1.2.2 for details);
- α_s: state-specific coefficient for the TFR effect;
- β_s: state-specific coefficient for the female education attainment effect;
- $\Theta_{d[s,t]}$: overall effect at the d-th knot matching from the proportion of ever-married females aged 15–19 for state-year $[s,t]$ (Section 2.2.1);
- $\Phi_{s,t}$: spatial-temporal effect, capturing spatial correlation at a certain time point and temporal correlation across spatial fields over time (Section 2.2.2).

α_s and β_s are the state-level random effects for TFR and female education attainment. They follow hierarchical normal distributions with mean at zero and a global precision parameter. For $s \in \{1, \cdots, k\}$, we have:

$$\alpha_s \sim \mathcal{N}(0, 1/\tau_\alpha), \tag{6}$$
$$\beta_s \sim \mathcal{N}(0, 1/\tau_\beta). \tag{7}$$

We assign non-informative priors to the global precision parameters τ_α and τ_β.

2.2.1 Model for Θ_d

$\Theta_{d[s,t]}$ models the overall non-linear relation between $\Psi_{s,t}$ and the state-level proportion of ever-married females aged 15–19 in Indian state/UT s, year t. We allow potential non-linear effect from this variable because it shows the strongest association (with the highest adjusted-like R^2 value) among all three variables against the ASFR 15–19 according to the exploratory plot in Figure 2. Index for $d[s,t]$ refers to matching the proportion of ever-married females aged 15–19 for state-year $[s,t]$ to the d-th knot, and is explained in details below.

Specifically, let $V_{s,t}$ denote the log of proportion of ever-married females aged 15–19 per 1,000 for Indian state/UT s in year t based on Bayesian model detailed in Section 1.2.1. We define a grid of values κ_d for $d \in \{1, \cdots, x\}$ where $\kappa_1 < \kappa_2 < \cdots < \kappa_x$ are the ordered and evenly distributed locations where the RW2 function evaluates $\boldsymbol{\Theta} = (\Theta_1, \cdots, \Theta_x)$. $x = 1000$ is the number of grid points where $\Theta_{d[s,t]}$ is evaluated. Let $\kappa_1 = \log(9.8)$, the minimum $V_{s,t}$, corresponding to 0.98% of the females aged 15–19 are married on the log scale. We set $\kappa_x = \log(938.9)$ as the maximal of $V_{s,t}$ across all state-years. We match each $V_{s,t}$ to the κ_d with the smallest absolute difference from $V_{s,t}$, denoting the d-th index for state-year $[s,t]$ as $d[s,t]$. Figure 4 illustrates how each $V_{s,t}$ is matched to a κ_d.

To model the relation between Θ_d and κ_d, we use a second-order random walk (RW2) process. In particular,

$$\Delta^2\Theta_d = \Theta_{d+2} - 2\Theta_{d+1} + \Theta_d, \text{ for } d \in \{1, \cdots, x-2\} \tag{8}$$
$$\Delta^2\Theta_d|\tau_\theta \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \tau_\theta^{-1}), \text{ for } d \in \{1, \cdots, x-2\}. \tag{9}$$

We assign a Penalized Complex (PC) prior to the regional precision parameter τ_θ:

$$\tau_\theta \sim \mathcal{PC}(z, 0.01), \tag{10}$$

where z is the standard deviation of y_i. The PC prior is a vague prior ([10] documented the PC prior specification in detail). Figure 5 illustrates the RW2 model results for Θ_d, the state-specific effects for the proportion of ever-married females aged 15–19 on ASFR, for ages 15–19. We limit the plot presentation to only show the effect within the range of proportions during 1990–2050 on the x-axis. As the model results suggest, the non-linear effect of state-specific proportion of ever-married females aged 15–19 on the ASFR 15–19 is different across Indian states/UTs and the non-linear effect is evident in most states/UTs.

2.2.2 Model for $\Phi_{s,t}$

$\Phi_{s,t}$ models the spatio-temporaleffects in Indian state/UT s, year t. It is modeled with a Besag-type intrinsic conditional autoregressive (ICAR) model [12] where (i) spatial correlation via a Besag spatial model across neighbouring Indian states/UTs; and (ii) an AR(1) model for the spatial fields over time.

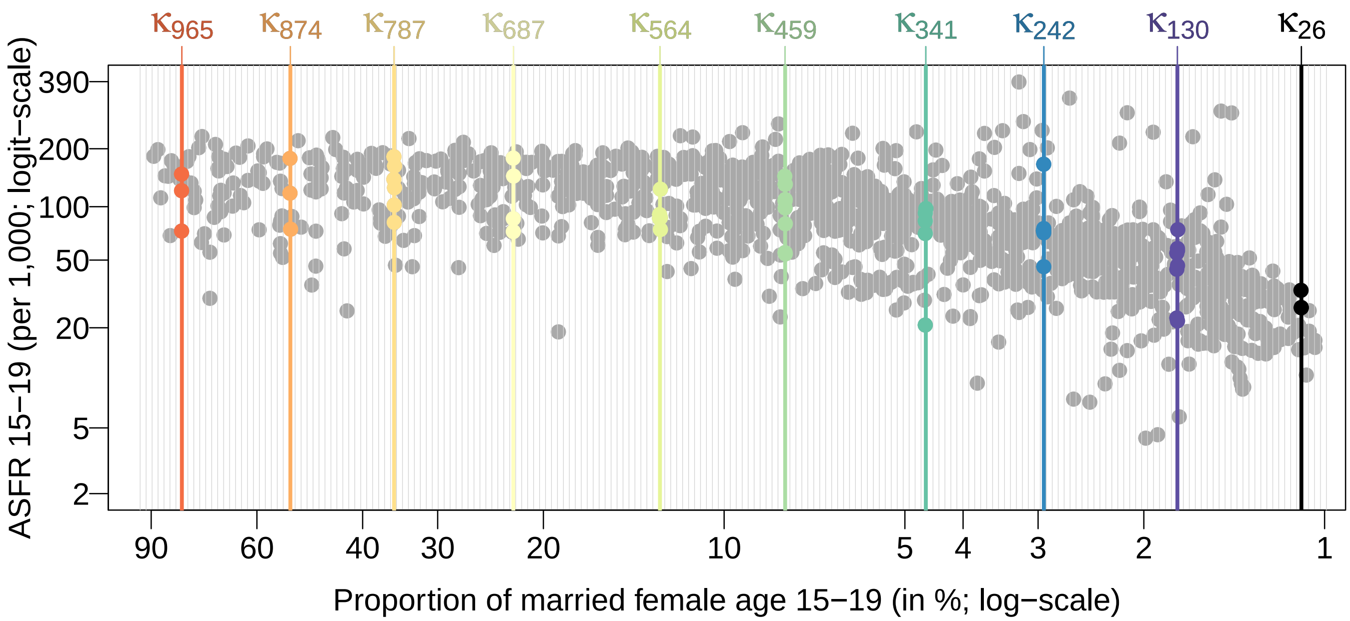

Figure 4: **RW2 index illustration.** Observed logit of ASFR 15–19 y_i against log of proportion of married female age 15–19 $V_{s,t}$ across all state-years are in grey dots. 200 vertical grey lines indicate one of every five knots κ_d for $d \in \{1, \cdots, 1000\}$ where $\Theta_{d[s,t]}$ is evaluated by the RW2 function. Ten κ_d's are highlighted for illustration. Dots with the same color as the location of highlighted κ_d all set to κ_d when modeling.

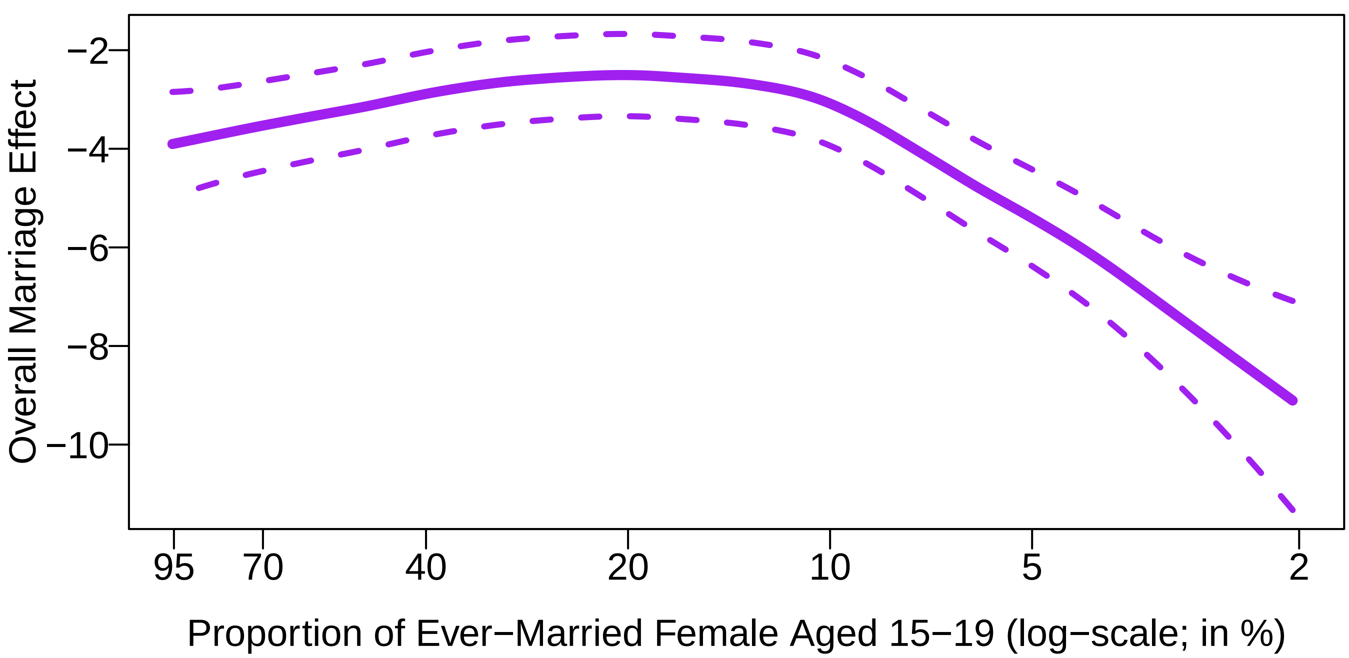

Figure 5: **Overall relation between proportion of ever-married females aged 15–19 and ASFR 15–19.** Median estimates for the proportion of ever-married females effect are in solid curve. 95% uncertainty intervals for model estimates are in dashed lines.

Spatial correlation in year t For a certain year t, we have $\Phi_t = (\Phi_{1,t}, \cdots, \Phi_{k,t})^\top$. The spatial structure is defined with respect to the Indian state/UT adjacency graph. Let $r \sim s$ indicate that Indian states/UTs r and s are neighbors (defined as sharing state/UT borders), and let n_s denote the number of neighbors that Indian state/UT s has. Then, conditional on the remaining states at time t, the Besag model is given by

$$\Phi_{s,t} \mid \{\Phi_{r,t} : r \neq s\}, \tau_\Phi \sim \mathcal{N}\left(\frac{1}{n_s}\sum_{r\sim s}\Phi_{r,t}, \frac{1}{n_s\tau_\Phi}\right). \tag{11}$$

The spatial correlation structure induces local spatial smoothing by assuming neighbouring Indian states/UTs (i.e. sharing boarders) have similar effects in year t. Figure 6 presents the posterior means of $\Phi_{s,t}$. We assign a PC vague prior to the precision parameter τ_Φ.

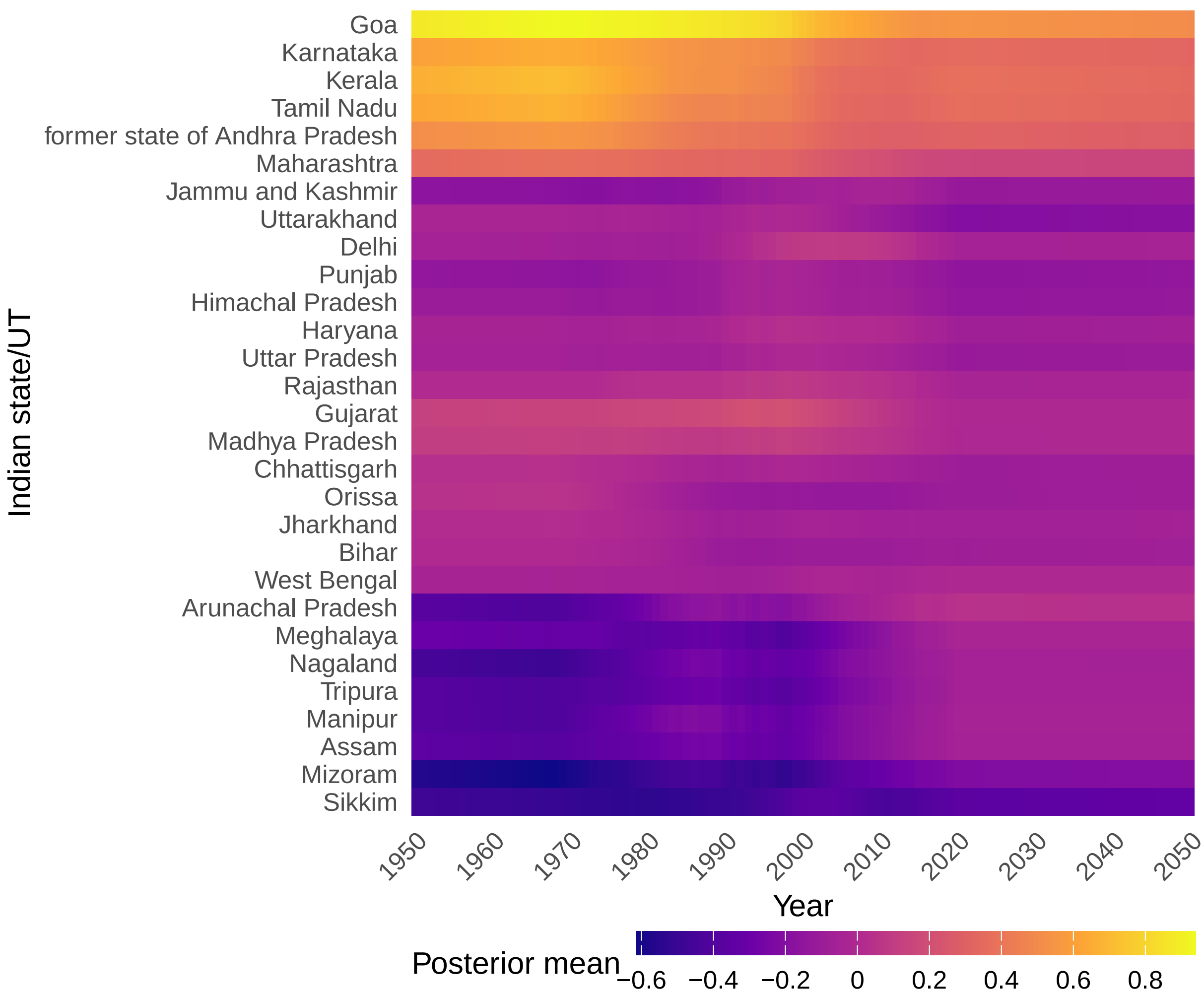

Figure 6: $\Phi_{s,t}$ **posterior mean.** Indian states/UTs are ordered based on their adjacency graph.

Temporal correlation across year t The sequence of spatial fields $\Phi_1, \ldots, \Phi_T$ is linked via an AR(1) process:

$$\Phi_t = \rho\Phi_{t-1} + \mathbf{u}_t, \text{ for } t \in \{2, ..., j\}, \tag{12}$$

where $|\rho| < 1$ is the temporal autocorrelation parameter and $\mathbf{u}_t$ is a spatial innovation vector. Hence, the spatial surface Φ_t evolves over time with persistence controlled by ρ. The temporal correlation parameter ρ was assigned the default INLA prior, which is a Gaussian prior on the transformed scale $\theta = \log\left(\frac{1+\rho}{1-\rho}\right)$ with $\theta \sim \mathcal{N}(0, 0.15^{-1})$.

2.3 Statistical computation

Computing of state-specific ASFR model We use the Integrated Nested Laplace Approximation (INLA) for Bayesian inference [13] of the state-level ASFR 15–19. We use the `R`-package **INLA** [11] to implement the INLA estimation procedure.

Computing of state-level female mean education years & proportion of ever-married females models We obtained posterior samples of all the model parameters and hyperparameters using a Markov chain Monte Carlo (MCMC) algorithm, implemented in the open source softwares `R 4.4.0` [14] and `JAGS 4.3.2` [15] (Just Another Gibbs Sampler), using `R`-packages **R2jags** [16] and **rjags** [17]. Results were obtained from 8 chains with a total number of 1,000 iterations in each chain, while the first 2,000 iterations were discarded as burn-in. After discarding burn-in iterations and proper thinning, the final posterior sample size for each parameter is 8,000. Convergence of the MCMC algorithm and the sufficiency of the number of samples obtained were checked through visual inspection of trace plots and convergence diagnostics of Gelman and Rubin [18], implemented in the **coda** `R`-package [19].

2.4 Constructing uncertainty intervals and point estimates

Throughout the project, we presented our results in the format of "point estimate [lower bound; upper bound]". E.g. the ASFR model results are presented as $Q^{\text{P.E.}}_{a,c,t}[Q^{\text{L}}_{a,c,t};Q^{\text{U}}_{a,c,t}]$. This section will explain how we derived each component in the result.

The uncertainties from the covariates were incorporated into the final uncertainties of the ASFR. For the g-th trajectory of the ASFR for age 15–19 on the logit scale $\Psi^{(g)}_{c,t}$, with $g \in \{1,\cdots,G\}$, it is obtained as:

$$\Psi^{(g)}_{s,t} = \alpha^{(g)}_s z^{(g)}_{s,t} + \beta^{(g)}_s x^{(g)}_{s,t} + \Theta^{(g)}_{d[s,t]} + \Phi^{(g)}_{s,t},$$

where $\alpha^{(g)}_s$, $\beta^{(g)}_s$, and $\Phi^{(g)}_{s,t}$ are the g-th trajectories of the model parameters α_s, β_s, and $\Phi_{s,t}$, obtained directly from the Bayesian model itself as posterior samples. $z^{(g)}_{s,t}$ is the g-th trajectory of the TFR for Indian state/UT s in year t on log scale (obtained in Section 1.2.3). $x^{(g)}_{s,t}$ is the g-th trajectory of the mean education years among all females aged 15–49 for Indian state/UT s in year t on log scale (obtained in Section 1.2.2). $\Theta^{(g)}_{d[s,t]}$ is the g-th trajectory of the overall effect based on the g-th trajectory of the proportion of ever-married females aged 15–19 $V^{(g)}_{c,t}$ (obtained in Section 1.2.1).

The 95% uncertainty intervals for $\Psi_{s,t}$ and $Q_{s,t}$, denoted as $[*^{\text{L}};*^{\text{U}}]$, are the 2.5th and 97.5th percentiles of the corresponding posterior samples. The point estimates $\Psi^{\text{P.E.}}_{s,t}$ and $Q^{\text{P.E.}}_{s,t}$ are the posterior medians:

$$\begin{aligned} *^{\text{L}} &= \text{percentile}_{2.5\%}\left\{*^{(1)},\ldots,*^{(G)}\right\}, \\ *^{\text{U}} &= \text{percentile}_{97.5\%}\left\{*^{(1)},\ldots,*^{(G)}\right\}, \\ *^{\text{P.E.}} &= \text{percentile}_{50\%}\left\{*^{(1)},\ldots,*^{(G)}\right\}, \end{aligned}$$

where $*$ can be substituted with $\Psi_{s,t}$ and $Q_{s,t}$.

3 Validation exercise and results

We validate the model performance by leaving out data collected after 2016 (equivalent to leaving out all data from NFHS 2019–21), rather than randomly leaving out data. This validation approach has been used in previous global health and population studies [20, 21, 22, 23, 24, 25]. We left out 261 observations, corresponding to 17.3% of the total observations (due to the varying number of observations collected in different survey years, the left-out observations can deviate from 20%). We call the left-out observations "testing dataset" and the remaining observations "training dataset". The validation results were calculated for 1,000 sets of left-out observations, where each set consisted of only one randomly selected left-out observation from each Indian state/UT. The reported validation results were based on the average of the outcomes from the 1,000 sets of left-out observations and the complete testing dataset for the validation exercise.

For each left-out observation in the testing dataset, we generate a posterior predictive distribution based on the Bayesian model fittings for the training dataset. We compute the error as the difference between the left-out observation and the median of the posterior predictive distribution. We report the median errors of all the left-out observations and the median of the absolute errors. We also report the coverage of the 95% prediction intervals (PIs). The lower and upper bounds of the 90% PI for each left-out observation is the 5th and 95th percentiles of the posterior predictive distribution. To summarize the coverage, we compute the proportion of left-out observations that fall outside the 90% in Table 4. Median errors and median absolute errors are close to zero for left-out observations. Coverage of 90% prediction intervals is higher than expected.

We also validate the model performance by comparing the model estimates based on the full and training datasets. We compute the error for each state-year as the difference between the median estimates based on the full dataset and those based on the training dataset. We report the median of these errors and absolute errors for all the state-years. We calculate the percentage of the state-years where the median estimates based on the full dataset fall outside the credible bounds of the estimates based on the training dataset. Table 5 shows the results for the comparison between estimates obtained based on the full dataset and estimates based on the training set. Median errors and the median absolute errors were close to zero. The proportion of updated estimates that fell outside the uncertainty intervals constructed based on the training set was small.

	All left-out obs. (1 complete set)	1 left-out obs. per state (1,000 sets average)
Median error	0.02	0.02
Median absolute error	0.02	0.01
% of left-out observations below 90% prediction interval	0.0	0.0
% of left-out observations above 90% prediction interval	0.0	0.0
Expected proportions(%)	**5**	**5**
# left out observations	261	29
% of left out to total observations	17.3%	–

Table 4: **Validation results for testing dataset.** Errors are defined as the difference between a left-out observation and the posterior median of its predictive distribution. Obs.: observation(s).

Year	**2000**	**2005**	**2010**	**2015**
Median Error	-0.00	-0.00	-0.00	-0.00
Median absolute error	0.00	0.00	0.00	0.00
Below 90% uncertainty interval (%)	0.0	0.0	0.0	0.0
Above 90% uncertainty interval (%)	0.0	0.0	0.0	0.0
Expected proportions(%)	$\leq$**5**	$\leq$**5**	$\leq$**5**	$\leq$**5**

Table 5: **Validation results for model run based on training dataset.** Errors are the differences between estimates based on the full dataset and the training set. Proportions are full run estimates that fall below or above 90% uncertainty interval of the validation run. The numbers of states/UTs fall above/below the uncertainty intervals are within the brackets after the proportions.

4 Model summary

Notation summary Table 6 summarizes the notations and indexes used for the ASFR 15–19 model. The notations for obtaining the covariates are not listed here.

Table 6: Notation summary.

Symbol	Description
	Indexes
i	Indicator for observations across all state-years, $i \in \{1, \cdots, n\}$, where $n = 1510$.
t	Indicator for year, $t \in \{1, \cdots, j\}$, where $j = 61$. $t = 1$ corresponds to the year 1990 and $t = j$ to the year 2050.
s	Indicator for state, $s \in \{1, \cdots, k\}$, where $k = 29$. We only present results in 21 states/UTs even though we model for all the 29 states/UTs.
p	Indicator for data source type, $p \in \{1, 2\}$. $p = 1$ corresponds to data source type NFHS, $p = 2$ to SRS.
d	Indicator for unique values of log-scaled proportion of ever-married females aged 15–19 $V_{s,t}$, $d \in \{1, \cdots, 1000\}$, where $d = 1$ corresponds to 9.75% and $d = 1000$ to 93.89%.
	Unknown parameters
$\Psi_{s,t}$	The true logit of ASFR aged 15–19 for Indian state/UT s, year t.
α_s	State-specific coefficient for the TFR effect.
β_s	State-specific coefficient for the female education attainment effect.
Θ_d	The overall relation between $\Psi_{s,t}$ and log-scaled proportion of ever-married females aged 15–19 $V_{s,t}$.
$\Phi_{s,t}$	The temporal correlation within state discrepancy between state-specific observations.
γ_s	State-specific drift for the temporal correlation process $\Phi_{s,t}$.
τ_ω	The precision parameter for non-sampling error variance.
τ_α	The precision parameter for α_s.
τ_β	The precision parameter for β_s.
τ_γ	The precision parameter for γ_s.
τ_θ	The precision parameter for the second-order increment $\Delta^2\Theta_d$.
$\tau_{\phi,s}$	The state-specific precision parameter for the first-order increment $\Delta\Phi_{s,t}$ for Indian state/UT s.
	Known quantities
y_i	i-th observed ASFR aged 15–19 for state $s[i]$ in year $t[i]$ on the logit scale.
u_i^2	The i-th sampling error variance for y_i.
z	The standard deviation of all y_i.
$x_{s,t}$	The mean education years among all females aged 15–49 for Indian state/UT s in year t.
$z_{s,t}$	The TFR for Indian state/UT s in year t.
$V_{s,t}$	The log of proportion of ever-married females aged 15–19 in Indian state/UT s in year t.
$\kappa_1, \cdots, \kappa_x$	Unique values of $V_{s,t}$ where Θ_d is evaluated.

Model for state-level age-specific fertility rate 15–19

$$
\begin{aligned}
y_i &= \Psi_{s[i],t[i]} + \Sigma_i, \text{ for } \forall i, \\
\Sigma_i | \lambda_p, \omega_p &\sim \mathscr{N}(\lambda_{p[i]}, u_i^2 + \omega_{p[i]}^2), \text{ for } \forall i, \\
\lambda_p &\sim \mathscr{N}(0, 1/\tau_\lambda), \text{ for } p \in \{1,2\}, \\
\Psi_{s,t} &= \alpha_s z_{s,t} + \beta_s x_{s,t} + \Theta_{d[s,t]} + \Phi_{s,t}, \text{ for } \forall s, \forall t, \\
\alpha_s &\sim \mathscr{N}(0, 1/\tau_\alpha), \text{ for } \forall s, \\
\beta_s &\sim \mathscr{N}(0, 1/\tau_\beta), \text{ for } \forall s, \\
\Delta^2 \Theta_d &= \Theta_{d+2} - 2\Theta_{d+1} + \Theta_d, \text{ for } d \in \{1, \cdots, x-2\}, \\
\Delta^2 \Theta_d | \tau_\theta &\overset{\text{i.i.d.}}{\sim} \mathscr{N}(0, \tau_\theta^{-1}), \text{ for } d \in \{1, \cdots, x-2\}, \\
\Phi_t &= (\Phi_{1,t}, \cdots, \Phi_{k,t})^\top, \\
\Phi_{s,t} \mid \{\Phi_{r,t} : r \neq s\}, \tau_\Phi &\sim \mathscr{N}\left(\frac{1}{n_s} \sum_{r \sim s} \Phi_{r,t}, \frac{1}{n_s \tau_\Phi}\right), \\
\Phi_t &= \rho \Phi_{t-1} + \mathbf{u}_t, \text{ for } t \in \{2, ..., j\}, \\
\log\left(\frac{1+\rho}{1-\rho}\right) = \theta &\sim \mathscr{N}(0, 0.15^{-1}), \\
\omega_p^{-2} &\sim \mathscr{PC}(z, 0.01), \text{ for } p \in \{1,2\} \\
\tau_\lambda &\sim \mathscr{G}(1, 0.00005), \\
\tau_\alpha &\sim \mathscr{G}(1, 0.00005), \\
\tau_\beta &\sim \mathscr{G}(1, 0.00005), \\
\tau_\gamma &\sim \mathscr{G}(1, 0.00005), \\
\tau_\Phi &\sim \mathscr{PC}(z, 0.01), \\
\tau_\theta &\sim \mathscr{PC}(z, 0.01).
\end{aligned}
$$

where $z = 0.61$ is the standard deviation of all the observations on the logit-scale.

Model for state-level proportion of ever-married females aged 15–19

$$
\begin{aligned}
v_j &\sim \mathscr{N}(V_{s[j],t[j]}, \omega_v^2), \text{ for } j \in \{1, \cdots, 125\}, \\
\exp\{V_{s,t}\} &= \frac{\delta_s^{\text{mar}} \cdot \exp\{\lambda_s^{\text{mar}} \cdot \log(t) + \zeta_s^{\text{mar}}\}}{1 + \exp\{\lambda_s^{\text{mar}} \cdot \log(t) + \zeta_s^{\text{mar}}\}}, \text{ for } \forall s, \forall t, \\
\delta_s^{\text{mar}} &\sim \mathscr{N}\left(\mu_\delta^{\text{mar}}, (\sigma_\delta^{\text{mar}})^2\right), \text{ for } \forall s, \\
\lambda_s^{\text{mar}} &\sim \mathscr{N}_{(-\infty, \log(100))}\left(\mu_\lambda^{\text{mar}}, (\sigma_\lambda^{\text{mar}})^2\right), \text{ for } \forall s, \\
\zeta_s^{\text{mar}} &\sim \mathscr{N}\left(\mu_\zeta^{\text{mar}}, \left(\sigma_\zeta^{\text{mar}}\right)^2\right), \text{ for } \forall s \\
\mu_\delta^{\text{mar}} &\sim \mathscr{U}(-0.5, 0.5), \\
\mu_\lambda^{\text{mar}} &\sim \mathscr{U}(-0.5, 0.5), \\
\mu_\zeta^{\text{mar}} &\sim \mathscr{U}(-0.5, 0.5), \\
\sigma_\delta^{\text{mar}} &\sim \mathscr{U}(0, 2), \\
\sigma_\lambda^{\text{mar}} &\sim \mathscr{U}(0, 2), \\
\sigma_\zeta^{\text{mar}} &\sim \mathscr{U}(0, 2), \\
\omega_v &\sim \mathscr{U}(0.001, 0.05).
\end{aligned}
$$

$\mathscr{U}(a, b)$ denotes a continuous uniform distribution with lower and upper bounds at a and b respectively.

Model for state-level mean education years in females aged 15–49

$$
\begin{aligned}
x_i &\sim \mathcal{N}(\chi_{s[i],t[i]}, \omega_\chi^2), \text{ for } i \in \{1, \cdots, 132\}, \\
\exp\{\chi_{s,t}\} &= \frac{\delta_s^{\text{edu}} \cdot \exp\left\{\lambda_s^{\text{edu}} \cdot \log(t) + \zeta_s^{\text{edu}}\right\}}{1 + \exp\{\lambda_s^{\text{edu}} \cdot \log(t) + \zeta_s^{\text{edu}}\}}, \text{ for } \forall s, \forall t, \\
\delta_s^{\text{edu}} &\sim \mathcal{N}\left(\mu_\delta^{\text{edu}}, (\sigma_\delta^{\text{edu}})^2\right), \text{ for } \forall s, \\
\lambda_s^{\text{edu}} &\sim \mathcal{N}_{(-\infty, \log(20))}\left(\mu_\lambda^{\text{edu}}, (\sigma_\lambda^{\text{edu}})^2\right), \text{ for } \forall s, \\
\zeta_s^{\text{edu}} &\sim \mathcal{N}\left(\mu_\zeta^{\text{edu}}, \left(\sigma_\zeta^{\text{edu}}\right)^2\right), \text{ for } \forall s, \\
\mu_\delta^{\text{edu}} &\sim \mathcal{U}(-0.5, 0.5), \\
\mu_\lambda^{\text{edu}} &\sim \mathcal{U}(-0.5, 0.5), \\
\mu_\zeta^{\text{edu}} &\sim \mathcal{U}(-0.5, 0.5), \\
\sigma_\delta^{\text{edu}} &\sim \mathcal{U}(0, 2), \\
\sigma_\lambda^{\text{edu}} &\sim \mathcal{U}(0, 2), \\
\sigma_\zeta^{\text{edu}} &\sim \mathcal{U}(0, 2), \\
\omega_\nu &\sim \mathcal{U}(0.001, 0.05).
\end{aligned}
$$

5 Sensitivity analyses

In addition to the validation exercises (Section 3), we conducted sensitivity analyses in this section to assess model assumptions. Essentially, we used two widely accepted criteria of Bayesian model assessment and comparison are Deviance Information Criterion (DIC) and Watanabe-Akaike Information Criterion (WAIC). In addition, we checked the cross-validation and predictive measures of fit using the Conditional Predictive Ordinates (CPO) to strengthen the model assessment and comparison.

5.1 Variable choice for nonparametric model

As presented in Section 2.2.1, a RW2 model is used to estimate the nonlinear effects Θ_d. The variable choice of using a nonparametric model on the proportion of ever-married females aged 15–19 instead of the other two variables (TFR and female education) may be considered as "inconsistent" with the empirical variable plot (Figure 2). As we explained in Section 1.2, the empirical trend can only be served as variable choice but not sufficient to imply the type of relation (linear or nonlinear).

We further tested the other two models where the RW2 process is applied to either TFR or mean education years for females aged 15–49, while assuming the other two variables linear. We compared the DIC, WAIC, and CPO for the two models in Table 7. A model with smaller DIC or WAIC should be preferred to one with larger DIC or WAIC. Whereas for CPO, a larger value indicates a model with better fit than one with a smaller CPO. All the three model performance indicators suggest the original model is better than the two alternative models.

Model	**DIC**	**WAIC**	**CPO**
Original	687	763	-400
Alternative 1	839	922	-475
Alternative 2	720	791	-418

Table 7: **Model comparison for time series setup.** CPO= $\sum_{i=1}^{n} \log(\text{cpo}_i)$. Original model: RW2 for female marriage proportion, as described in Section 2.2. Alternative 1 model: RW2 for TFR. Alternative 2 model: RW2 for female education.

In addition to the better model fit for the original model than the two alternative models, there are two other reasons we chose the original model against the alternative models:

1. Figure 7 top: the original model provides more reasonable uncertainty bounds to cover more data points than the alternative model 1 (RW2 on TFR). The uncertainty bounds from the original model better reflect the drastic differences in the fertility patterns from difference data sources.
2. Figure 7 bottom: the original model provides narrower and more reasonable uncertainty bounds for backward projections and with wider and more reasonable uncertainty bounds for forward projections than the alternative model.

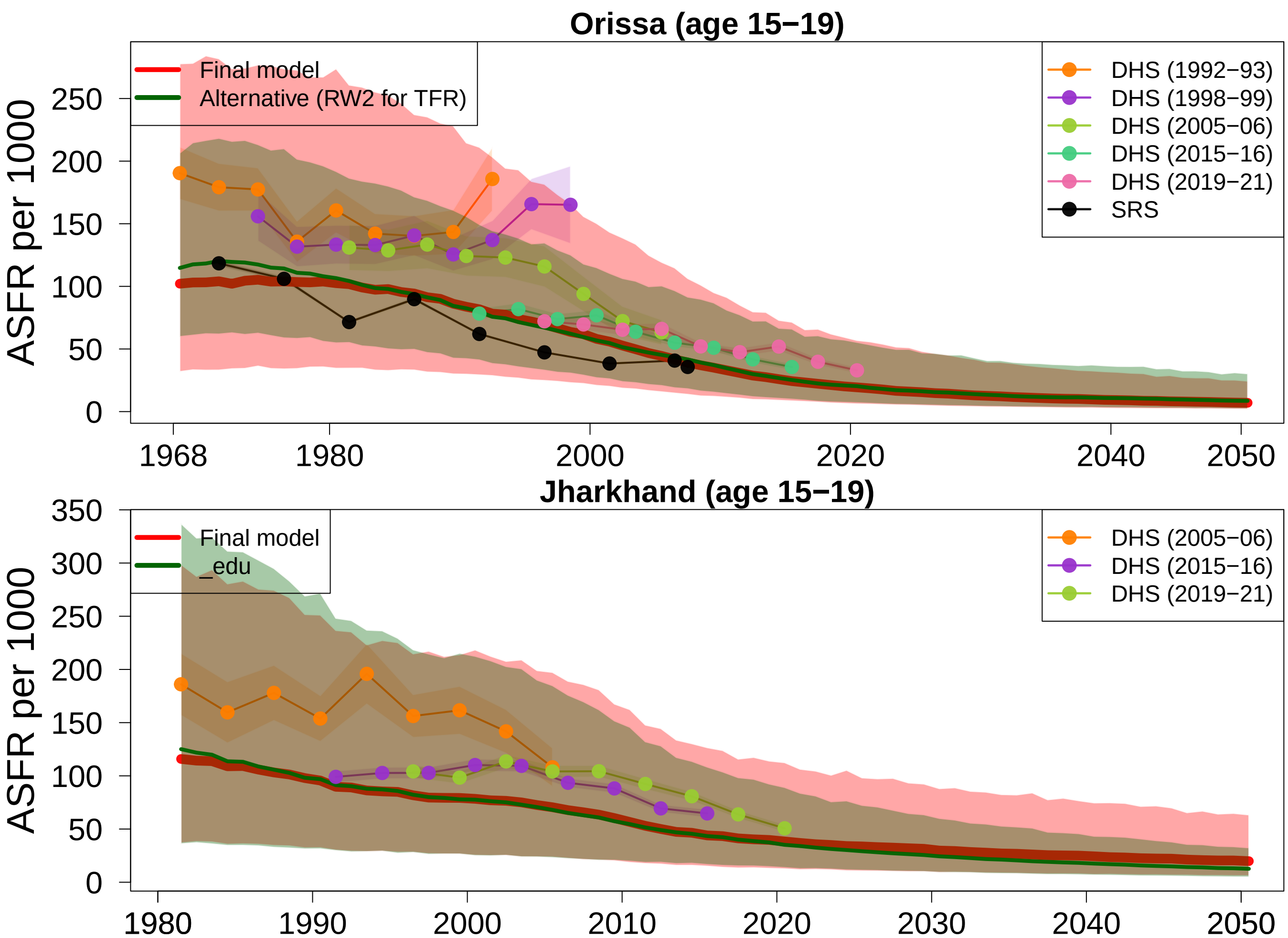

Figure 7: **ASFR 15–19 model fit comparison during 1950–2023.** Top: compare with alternative model 1 with RW2 on TFR. Bottom: compare with alternative model 2 with RW2 on female education years. Red curves and shades: median estimates and 95% uncertainty intervals from the original model (Section 2.2). Green curves and shades: median estimates and 95% uncertainty intervals from the alternative model. Dots with connection lines: input data series differentiated by colors. Shades around the data series: sampling errors or stochastic errors.

5.2 Model choice for the nonlinear element

Section 5.1 concludes that applying a nonlinear structure to the proportion of ever-married females aged 15–19 achieves the best model fit. This section explains why we choose RW2 instead of other nonlinear models such as splines, or higher order polynomials.

To test this model assumption, we conducted model diagnostics and checks among four models:

- Original model: use RW2 for ever-married female proportion (Section 2.2.1).

- Alternative model 1: use cubic B-splines for ever-married female proportion while keeping all the other model settings unchanged.
- Alternative model 2: use order polynomials (order 4) for ever-married female proportion while keeping all the other model settings unchanged.
  - We fitted one with order 3 $v^3_{s,t}$ initially.
  - Model performance indicators in Table 8 shows the one with order 4 $v^4_{s,t}$.

We compared the DIC, WAIC, and the CPO for the two models in Table 8. All three criteria unanimously identify the original model out performs the three alternative models.

Model 10–14	**DIC**	**WAIC**	**CPO**
Original	720	791	-418
Alternative 1	1192	1186	-879
Alternative 2	1204	1271	-656

Table 8: **Model comparison for Θ nonlinear setup.** CPO= $\sum_{i=1}^{n}\log(\text{cpo}_i)$. Original model: use RW2 for Θ (Section 2.2.1). Alternative model 1: use cubic B-splines for Θ. Alternative model 2: use polynomial of order 4 for Θ.

Conventionally, a spline regression function or higher order polynomials is used to model non-linear relationship. Instead, the original model uses a RW2 model as a continuous time process [26] for the overall relation $\boldsymbol{\Theta}$. The approach focuses on approximating some continuous spline on a discrete set of knots $(\kappa_1, \cdots, \kappa_x)$. There are several key advantages of using RW2 process in the Integrated Nested Laplace Approximation (INLA) framework [13]:

- The splines approach is restricted to choosing the appropriate number of locations of knots [27]. An RW2 model on a regular grid results in a very sparse precision matrix, which INLA can compute efficiently [26].
- RW2 is locally more adaptive and it automatically penalizes for complexity within the Bayesian INLA framework [26]. By contrast, polynomials can only increase flexibility by manually adding higher-order terms, which risks overfitting that are not justified by the data.
- Estimation via splines can be computationally costly. The smoothing parameter in splines needs to be specified and estimated. This could result in slow convergence and costly MCMC sampling. In contrast, RW2 has a natural penalty structure which fits naturally into the INLA framework.

5.3 Prior choice for main model elements

To assess the robustness of the projected estimates to prior assumptions on hierarchical variance components, we performed a structured sensitivity analysis for the hyperpriors assigned to the main random effects in the model. This analysis was motivated by the possibility that, in hierarchical models, prior choices for hyperparameters can influence posterior smoothing, the degree of borrowing across units, and ultimately projected trajectories, especially in settings with limited data information.

The sensitivity analysis focused on the hyperpriors for the precision parameters of the three main components that are most relevant to smoothing and borrowing information:

- Element one: the state-specific random effects for data source type;
- Element two: the overall RW2 effect for proportion of ever-married females aged 15–19; and
- Element three: the spatial Besag effect for states/UTs, with temporal dependence introduced through an AR1 time series process.

5.3.1 Prior scenarios

In the final model, the precision hyperparameters for these components were assigned PC priors of the form:

$$P(\sigma > z) = \alpha,$$

with z the SD of all y_i's and $\alpha = 0.01$. σ denotes the marginal standard deviation of the latent effect and $\tau = 1/\sigma^2$ is the corresponding precision. Under the baseline prior, 1% prior probability is assigned to latent standard deviations exceeding z.

To evaluate sensitivity to the degree of prior shrinkage, we considered alternative PC priors that were either more informative or more vague than the baseline while preserving the same structural form and keeping $\alpha = 0.01$ fixed. For each component, we used two alternative priors as listed in Table 9. These choices represent a two-fold decrease or increase in the upper-scale parameter z relative to the baseline. A smaller value at $z/2$ induces stronger shrinkage toward lower latent variability, whereas a larger value at $2z$ allows greater variability in the corresponding random effect. Together with the baseline model, two alternative prior choices on three model elements yielded a total of nine prior scenarios.

Prior scenario	**PC prior setting**
Baseline	$P(\sigma > z) = 0.01$
More vague	$P(\sigma > 2z) = 0.01$
More informative	$P(\sigma > z/2) = 0.01$

Table 9: Alternative prior specifications in the sensitivity analysis.

5.3.2 Model sensitivity by estimation and projection periods

We evaluated model sensitivity to the priors for the estimation period 1990–2025 and the projection period 2026–2050. In each period, we summarized the sensitivity measurements by two sets of Indian states/UTs based on their data coverage, because the influence of hyperpriors is expected to depend on the amount (in quality and quantity) of information available in the data. We calculated the state-specific data coverage as #observations$/(2025-1990+1)$. That is, the number of observations per estimation year. The state groupings based on data coverage are:

- Data-poor state (data coverage $< 70\%$): Chhattisgarh, Jharkhand, Uttarakhand.
- Data-rich state (data coverage $> 90\%$): all remaining states/UTs.

Table 10 presents the measurements comparing the alternative models to the baseline model via: (i) changing a single prior component one at a time; and (ii) simultaneous changes in multiple priors. The following measurements are calculated for each state grouping (data-poor, data-rich, and all states combined), each prior scenario (nine in total), and each period (1990–2025 and 2026–2050):

1. Difference in model estimates: trajectories of ASFR from alternative models minus trajectories of ASFR from the baseline (final) model.
2. CI width ratio: 95% credible interval width of ASFR from alternative models divided by the 95% credible interval width of ASFR from the baseline (final) model.

First, the model is robust during the projection period for all prior scenarios and is subject to certain level of sensitivity during the estimation period under one or two prior scenarios. The difference in ASFR between the alternative models and the baseline (final) model are close to zero and not significantly different from zero for all scenarios in both the estimation period 1990–2025 and the projection period 2026–2050. The only few exceptions in the estimation period are when we assign more vague priors to the precision parameters to the data source type ("type vague") which is a main contributor for the sensitivity under the scenario where all hyperpriors are assigned with more vague priors ("all vague"). In these two vague prior scenarios, the alternative models tend to project larger ASFR than by the baseline (final) model, but the effect is statistically insignificant. Figure 8 shows the model results comparison for the two identified prior scenarios with relative large difference in estimation period from the baseline (final) model. For each prior scenario, we present the state results with the largest average difference in ASFR for a certain scenario to illustration the most extreme case. Based on Figure 8, even the most outlying state results between the two prior scenarios are negligible.

Second, the credible intervals of the results tend to be wider than the final model under the scenario with more vague prior on data source type precision parameter ("type vague") and on all precision parameters ("all vague"). On the other hand, using more informative priors on all precision parameters ("all informative") produce narrower

Table 10: **Summary of prior sensitivity analysis by estimation and projection periods, for groups of states based on data availability.** Difference: ASFR (alternative model minus final model). CI width ratio: 95% CI width (alternative model divided by final model). All measurements are the averages over selected states/UTs during All years in period Year 1–Year 2. †: measurement is significantly different from zero.

Prior scenario	State/UT	Difference (per 1,000) 1990–2025	Difference (per 1,000) 2026–2050	CI width ratio 1990–2025	CI width ratio 2026–2050
Baseline	–	0.0	0.0	1.00	1.00
all vague	All	2.6	1.0	1.21	1.20
	All	[-7.4; 11.1]	[-4.7; 5.3]		
type vague	All	2.2	1.0	1.18	1.17
	All	[-7.7; 10.7]	[-4.8; 5.2]		
rw2 vague	All	0.1	0.1	1.00	1.00
	All	[-9.9; 8.3]	[-5.5; 4.3]		
besag vague	All	0.7	0.3	1.06	1.06
	All	[-9.2; 9.0]	[-5.3; 4.5]		
type informative	All	-0.9	-0.3	0.91	0.93
	All	[-10.7; 7.3]	[-5.8; 3.9]		
rw2 informative	All	0.0	0.1	0.99	1.00
	All	[-9.9; 8.3]	[-5.4; 4.3]		
besag informative	All	-1.0	-0.1	0.91	0.92
	All	[-10.7; 7.3]	[-5.7; 4.0]		
all informative	All	-2.1	-0.5	0.82	0.84
	All	[-11.6; 6.2]	[-6.0; 3.6]		
all vague	Data-poor	4.8	2.3	1.19	1.20
	Data-poor	[-46.6; 29.1]	[-17.7; 12.0]		
type vague	Data-poor	4.4	2.0	1.16	1.17
	Data-poor	[-47.1; 28.6]	[-17.9; 11.9]		
rw2 vague	Data-poor	2.1	1.2	1.00	1.00
	Data-poor	[-48.5; 25.8]	[-19.0; 10.4]		
besag vague	Data-poor	3.0	1.6	1.06	1.06
	Data-poor	[-47.7; 26.8]	[-18.7; 10.9]		
type informative	Data-poor	1.2	0.7	0.92	0.92
	Data-poor	[-49.0; 24.6]	[-19.7; 9.9]		
rw2 informative	Data-poor	1.9	1.2	0.99	0.99
	Data-poor	[-48.6; 25.7]	[-19.0; 10.3]		
besag informative	Data-poor	0.8	0.6	0.92	0.92
	Data-poor	[-49.5; 24.5]	[-19.3; 9.7]		
all informative	Data-poor	-0.3	0.1	0.83	0.84
	Data-poor	[-50.2; 23.0]	[-20.0; 9.2]		
all vague	Data-rich	2.7	0.9	1.21	1.20
	Data-rich	[-6.9; 11.3]	[-4.6; 5.3]		
type vague	Data-rich	2.3	0.9	1.18	1.17
	Data-rich	[-7.2; 10.9]	[-4.6; 5.3]		
rw2 vague	Data-rich	0.2	0.2	1.00	1.00
	Data-rich	[-9.2; 8.5]	[-5.3; 4.4]		
besag vague	Data-rich	0.8	0.3	1.06	1.06
	Data-rich	[-8.6; 9.1]	[-5.2; 4.6]		
type informative	Data-rich	-0.8	-0.2	0.91	0.93
	Data-rich	[-10.1; 7.3]	[-5.7; 4.0]		
rw2 informative	Data-rich	0.1	0.2	0.99	1.00
	Data-rich	[-9.3; 8.4]	[-5.3; 4.4]		
besag informative	Data-rich	-0.8	0.0	0.91	0.92
	Data-rich	[-10.3; 7.5]	[-5.5; 4.2]		
all informative	Data-rich	-1.8	-0.3	0.82	0.84
	Data-rich	[-11.2; 6.2]	[-5.8; 3.8]		

credible intervals than the final model. Figure 9 illustrates the model results comparison for the three prior scenarios with relatively large changes (wider or narrower) in uncertainty intervals from the baseline (final) model. For each prior scenario, we present the state results with the average CI width ratio that is the furthest away from 1.00 (above or below 1) for a certain scenario. As shown in Figure 9, even the most extreme state results are not much affected by the prior choices in terms of uncertainty intervals.

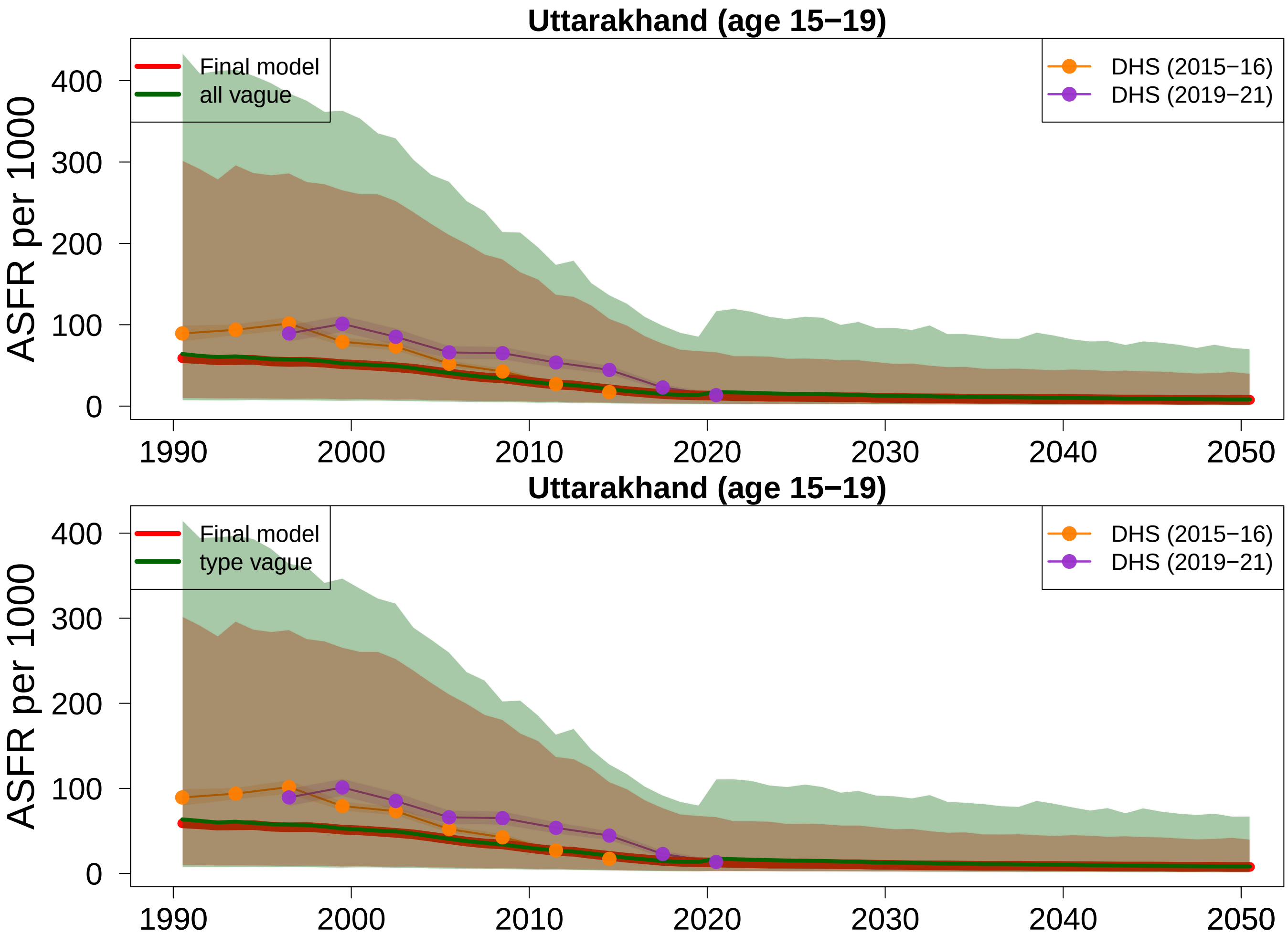

Figure 8: **ASFR 15–19 model comparison based on alternative prior scenarios from selected states/UTs with large changes in model estimates.** Top: compare with alternative model with more vague prior on all precision parameters. Bottom: compare with alternative model with more vague prior on precision parameter of data source types. Red curves and shades: median estimates and 95% uncertainty intervals from the final model (Section 2.2). Green curves and shades: median estimates and 95% uncertainty intervals from alternative models under varying prior scenarios. Dots with connection lines: input data series differentiated by colors. Shades around the data series: sampling errors or stochastic errors.

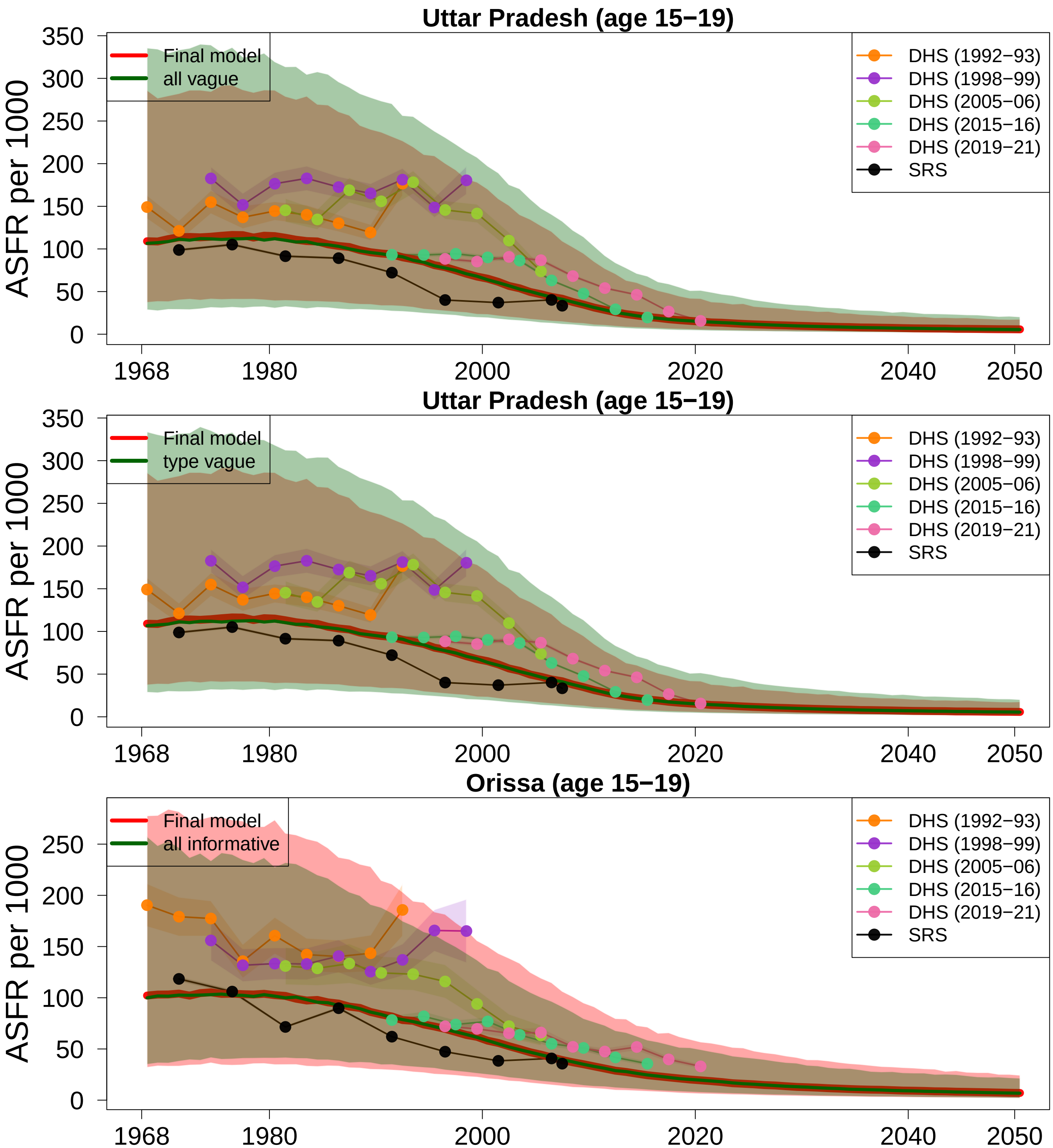

Figure 9: **ASFR 15–19 model comparison based on alternative prior scenarios from selected states/UTs with large changes in uncertainty intervals.** Top: compare with alternative model with more vague prior on all precision parameters. Middle: compare with alternative model with more vague prior on precision parameter of data source types. Bottom: compare with alternative model with more informative prior on all precision parameters. Red curves and shades: median estimates and 95% uncertainty intervals from the final model (Section 2.2). Green curves and shades: median estimates and 95% uncertainty intervals from alternative models under varying prior scenarios. Dots with connection lines: input data series differentiated by colors. Shades around the data series: sampling errors or stochastic errors.

6 GATHER Compliance

Guidelines for Accurate and Transparent Health Estimates Reporting (GATHER) are reported in Table 11.

Table 11: **GATHER checklist.** Checklist of information that should be included in new reports of global health estimates. M: main manuscript. A: Appendix. Section titles after "M" are stated for the main manuscript. Page numbers after "A" are shown for the appendix. GATHER: Guidelines for Accurate and Transparent Health Estimates Reporting

Item #	Checklist item	Reported on page # or section #
Objectives and funding		
1	Define the indicator(s), populations (including age, sex, and geographic entities), and time period(s) for which estimates were made.	M-Introduction, A1–1
2	List the funding sources for the work.	M-Abstract
Data Inputs		
For all data inputs from multiple sources that are synthesized as part of the study:		
3	Describe how the data were identified and how the data were accessed.	M-Database construction, A1–4
4	Specify the inclusion and exclusion criteria. Identify all ad-hoc exclusions.	M-Database construction, A4–4
5	Provide information on all included data sources and their main characteristics. For each data source used, report reference information or contact name/institution, the population represented, data collection method, year(s) of data collection, sex and age range, diagnostic criteria or measurement method, and sample size, as relevant.	A1–1 & A28–41
6	Identify and describe any categories of input data that have potentially important biases (e.g., based on characteristics listed in item 5).	A4–7
For data inputs that contribute to the analysis but were not synthesized as part of the study:		
7	Describe and give sources for any other data inputs.	M-Database construction, A28–41
For all data inputs:		
8	Provide all data inputs in a file format from which data can be efficiently extracted (e.g., a spreadsheet rather than a PDF), including all relevant meta-data listed in item 5. For any data inputs that cannot be shared because of ethical or legal reasons, such as third-party ownership, provide a contact name or the name of the institution that retains the right to the data.	M-Database construction
Data analysis		
9	Provide a conceptual overview of the data analysis method. A diagram may be helpful.	M-Database construction, A7
10	Provide a detailed description of all analysis steps, including mathematical formulae. This description should cover, as relevant, data cleaning, data pre-processing, data adjustments and weighting of data sources, and mathematical or statistical model(s).	M-Statistical analysis, A7–13
11	Describe how candidate models were evaluated and how the final model(s) were selected.	M-Statistical analysis, A8

Continued on next page

Table 11 – continued from previous page

Item #	Checklist item	Reported on page # or section #
12	Provide the results of an evaluation of model performance, if done, as well as the results of any relevant sensitivity analysis.	A8–13
13	Describe methods for calculating the uncertainty of the estimates. State which sources of uncertainty were, and were not, accounted for in the uncertainty analysis.	A12–14
14	State how analytic or statistical source code used to generate estimates can be accessed.	M-Statistical analysis
Results and Discussion		
15	Provide published estimates in a file format from which data can be efficiently extracted.	M-Results, A45-67
16	Report a quantitative measure of the uncertainty of the estimates (e.g., uncertainty intervals).	A13–14
17	Interpret results in light of existing evidence. If updating a previous set of estimates, describe the reasons for changes in estimates.	M-Results
18	Discuss limitations of the estimates. Include a discussion of any modelling assumptions or data limitations that affect the interpretation of the estimates.	M-Discussion

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7 Supplementary Tables

- Table 12: Data sources for age-specific fertility rate aged 15–19, by Indian state/UT (page 28–32).
- Table 13: Data sources for proportion of ever-married females aged 15–19, by Indian state/UT (page 33–36).
- Table 14: Data sources for mean education years among females aged 15–49, by Indian state/UT (page 37–40).

Table 12: **Data sources for age-specific fertility rate aged 15–19, by Indian state/UT.** For each Indian state/UT, the total number of observations and the most recent observation year are shown after the state/UT name. For each state-specific data series, the number of observations and the most recent observation year within that series are shown before each data series name. The source type that each data series falls in is shown in parentheses after each data series name.

Indian state/UT	# obs.	Most recent obs. year	Data series name [source type]
former state of Andhra Pradesh	54	2018.5	
	9	1991.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2014.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2018.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	9	2007.5	Sample Registration System [SRS Direct]
Arunachal Pradesh	63	2020.5	
	9	1992.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2015.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2020.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	18	2007.5	Sample Registration System [SRS Direct]
Assam	54	2018.5	
	9	1992.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2015.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2018.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	9	2007.5	Sample Registration System [SRS Direct]
Bihar	52	2018.5	
	9	1992.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2014.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2018.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	7	2007.5	Sample Registration System [SRS Direct]
Chhattisgarh	27	2020.5	
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2015.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2020.5	2019–2021 National Family Health Survey [Standard DHS Direct]
Delhi	45	2020.5	

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Table 12 – continued from previous page

Indian state/UT	# obs.	Most recent obs. year	Data series name [source type]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2015.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2020.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	18	2007.5	Sample Registration System [SRS Direct]
Goa	63	2018.5	
	9	1992.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2014.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2018.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	18	2007.5	Sample Registration System [SRS Direct]
Gujarat	54	2018.5	
	9	1992.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2015.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2018.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	9	2007.5	Sample Registration System [SRS Direct]
Haryana	54	2020.5	
	9	1992.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2014.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2020.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	9	2007.5	Sample Registration System [SRS Direct]
Himachal Pradesh	63	2018.5	
	9	1991.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2015.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2018.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	18	2007.5	Sample Registration System [SRS Direct]
Jammu and Kashmir	36	2019.5	
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2015.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2019.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	9	2007.5	Sample Registration System [SRS Direct]
Jharkhand	27	2020.5	
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2015.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2020.5	2019–2021 National Family Health Survey [Standard DHS Direct]
Karnataka	54	2018.5	
	9	1992.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2014.5	2015–2016 National Family Health Survey [Standard DHS Direct]

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Table 12 – continued from previous page

Indian state/UT	# obs.	Most recent obs. year	Data series name [source type]
	9	2018.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	9	2007.5	Sample Registration System [SRS Direct]
Kerala	54	2018.5	
	9	1992.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2015.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2018.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	9	2007.5	Sample Registration System [SRS Direct]
Madhya Pradesh	54	2020.5	
	9	1991.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2014.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2020.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	9	2007.5	Sample Registration System [SRS Direct]
Maharashtra	54	2018.5	
	9	1992.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2014.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2018.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	9	2007.5	Sample Registration System [SRS Direct]
Manipur	63	2018.5	
	9	1992.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2014.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2018.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	18	2007.5	Sample Registration System [SRS Direct]
Meghalaya	63	2018.5	
	9	1992.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2014.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2018.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	18	2007.5	Sample Registration System [SRS Direct]
Mizoram	57	2018.5	
	9	1992.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2015.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2018.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	12	2007.5	Sample Registration System [SRS Direct]
Nagaland	63	2018.5	
	9	1992.5	1992–1993 National Family Health Survey [Standard DHS Direct]

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Table 12 – continued from previous page

Indian state/UT	# obs.	Most recent obs. year	Data series name [source type]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2015.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2018.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	18	2007.5	Sample Registration System [SRS Direct]
Odisha	54	2020.5	
	9	1992.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2015.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2020.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	9	2007.5	Sample Registration System [SRS Direct]
Punjab	54	2020.5	
	9	1992.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2015.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2020.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	9	2007.5	Sample Registration System [SRS Direct]
Rajasthan	53	2020.5	
	9	1992.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2015.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2020.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	8	2007.5	Sample Registration System [SRS Direct]
Sikkim	54	2018.5	
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2014.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2018.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	18	2007.5	Sample Registration System [SRS Direct]
Tamil Nadu	54	2020.5	
	9	1991.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2014.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2020.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	9	2007.5	Sample Registration System [SRS Direct]
Tripura	63	2018.5	
	9	1992.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1999.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2014.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2018.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	18	2007.5	Sample Registration System [SRS Direct]

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Table 12 – continued from previous page

Indian state/UT	# obs.	Most recent obs. year	Data series name [source type]
Uttar Pradesh	54	2020.5	
	9	1992.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2015.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2020.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	9	2007.5	Sample Registration System [SRS Direct]
Uttarakhand	18	2020.5	
	9	2014.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2020.5	2019–2021 National Family Health Survey [Standard DHS Direct]
West Bengal	52	2018.5	
	9	1991.5	1992–1993 National Family Health Survey [Standard DHS Direct]
	9	1998.5	1998–1999 National Family Health Survey [Standard DHS Direct]
	9	2005.5	2005–2006 National Family Health Survey [Standard DHS Direct]
	9	2014.5	2015–2016 National Family Health Survey [Standard DHS Direct]
	9	2018.5	2019–2021 National Family Health Survey [Standard DHS Direct]
	7	2007.5	Sample Registration System [SRS Direct]

Table 13: **Data sources for ever-married females aged 15–19, by Indian state/UT.** For each Indian state/UT, the total number of observations and the most recent observation year are shown after the state/UT name. For each state-specific data series, the number of observations and the most recent observation year within that series are shown before each data series name. The source type that each data series falls in is shown in parentheses after each data series name.

Indian state/UT	# obs.	Most recent obs. year	Data series name [source type]
former state of Andhra Pradesh	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Arunachal Pradesh	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Assam	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Bihar	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Chhattisgarh	3	2020.4	
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
Delhi	3	2020.4	
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Goa	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Gujarat	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]

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Table 13 – continued from previous page

Indian state/UT	# obs.	Most recent obs. year	Data series name [source type]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Haryana	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Himachal Pradesh	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Jammu and Kashmir	3	2020.4	
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Jharkhand	3	2020.4	
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Karnataka	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Kerala	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Madhya Pradesh	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
Maharashtra	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Manipur	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]

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Table 13 – continued from previous page

Indian state/UT	# obs.	Most recent obs. year	Data series name [source type]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Meghalaya	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Mizoram	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Nagaland	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Odisha	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Punjab	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Rajasthan	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
Sikkim	4	2020.4	
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Tamil Nadu	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]

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Table 13 – continued from previous page

Indian state/UT	# obs.	Most recent obs. year	Data series name [source type]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Tripura	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Uttar Pradesh	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
Uttarakhand	2	2020.4	
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
West Bengal	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]

Table 14: **Data sources for mean education years among females aged 15–49, by Indian state/UT.** For each Indian state/UT, the total number of observations and the most recent observation year are shown after the state/UT name. For each state-specific data series, the number of observations and the most recent observation year within that series are shown before each data series name. The source type that each data series falls in is shown in parentheses after each data series name.

Indian state/UT	# obs.	Most recent obs. year	Data series name [source type]
former state of Andhra Pradesh	4	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Arunachal Pradesh	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Assam	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Bihar	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Chhattisgarh	3	2020.4	
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Delhi	3	2020.4	
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Goa	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Gujarat	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]

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Table 14 – continued from previous page

Indian state/UT	# obs.	Most recent obs. year	Data series name [source type]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Haryana	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Himachal Pradesh	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Jammu and Kashmir	3	2020.4	
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Jharkhand	3	2020.4	
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Karnataka	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Kerala	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Madhya Pradesh	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Maharashtra	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]

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Table 14 – continued from previous page

Indian state/UT	# obs.	Most recent obs. year	Data series name [source type]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Manipur	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Meghalaya	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Mizoram	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Nagaland	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Odisha	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Punjab	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Rajasthan	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Sikkim	4	2020.4	
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]

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Table 14 – continued from previous page

Indian state/UT	# obs.	Most recent obs. year	Data series name [source type]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Tamil Nadu	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Tripura	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Uttar Pradesh	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
Uttarakhand	2	2020.4	
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]
West Bengal	5	2020.4	
	1	1993	1992–1993 National Family Health Survey [Standard DHS Direct]
	1	1999.4	1998–1999 National Family Health Survey [Standard DHS Direct]
	1	2006.2	2005–2006 National Family Health Survey [Standard DHS Direct]
	1	2016	2015–2016 National Family Health Survey [Standard DHS Direct]
	1	2020.4	2019–2021 National Family Health Survey [Standard DHS Direct]

8 Supplementary Figures

- Figure 10: Proportion of ever-married females aged 15–19 by Indian state/UT model results.
- Figure 11: Mean education years for females aged 15–49 by Indian state/UT model results.
- Figure 12: Total fertility rate by Indian state/UT model results.
- Figure 13: ASFR age 15–19 estimates and projections by Indian state/UT, 1990–2050 (page 45–67).

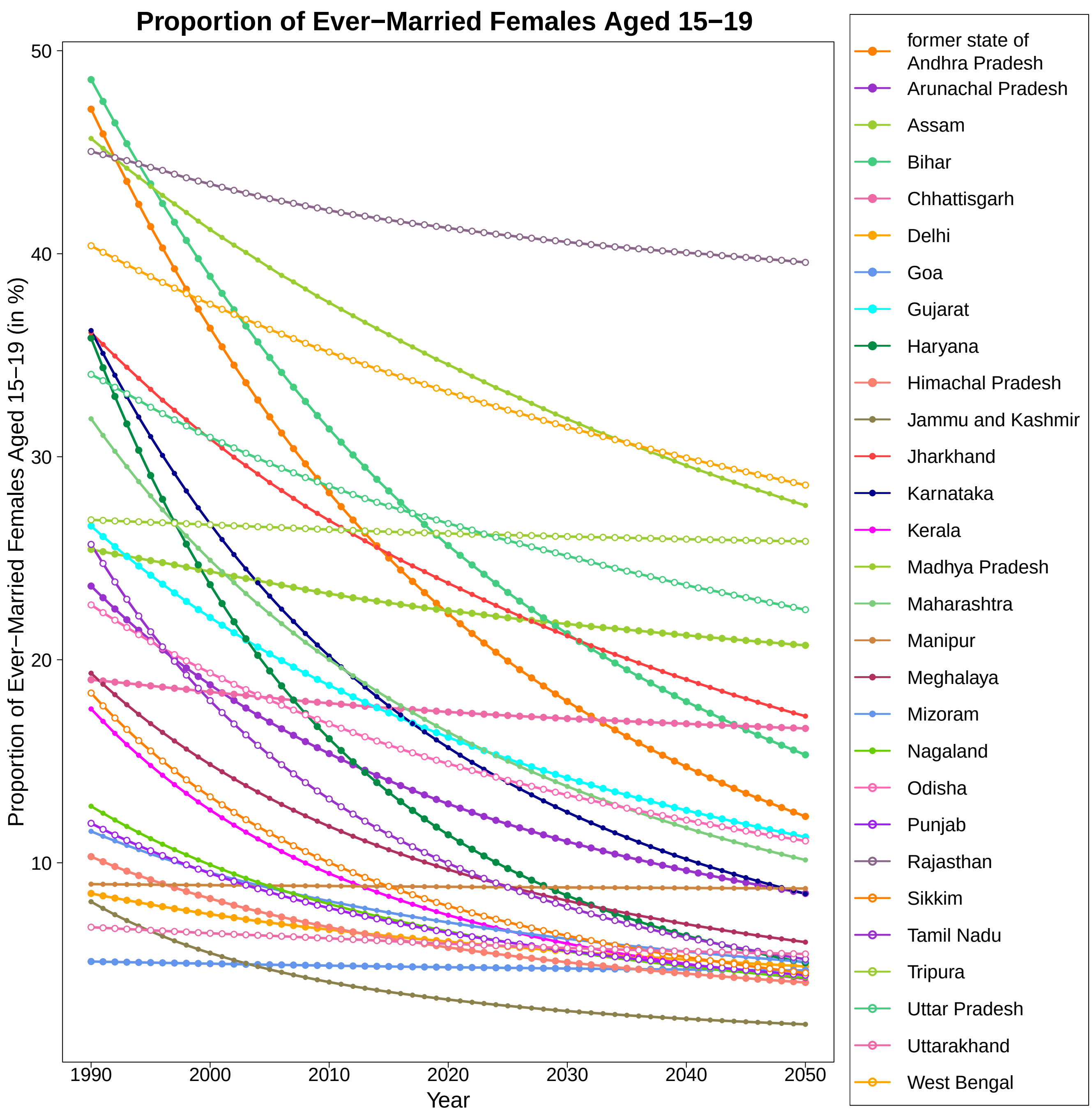


Figure 10: **Proportion of ever-married females aged 15–19 by Indian state/UT model results.** Model medians are shown. States/UTs are differentiated by colors.

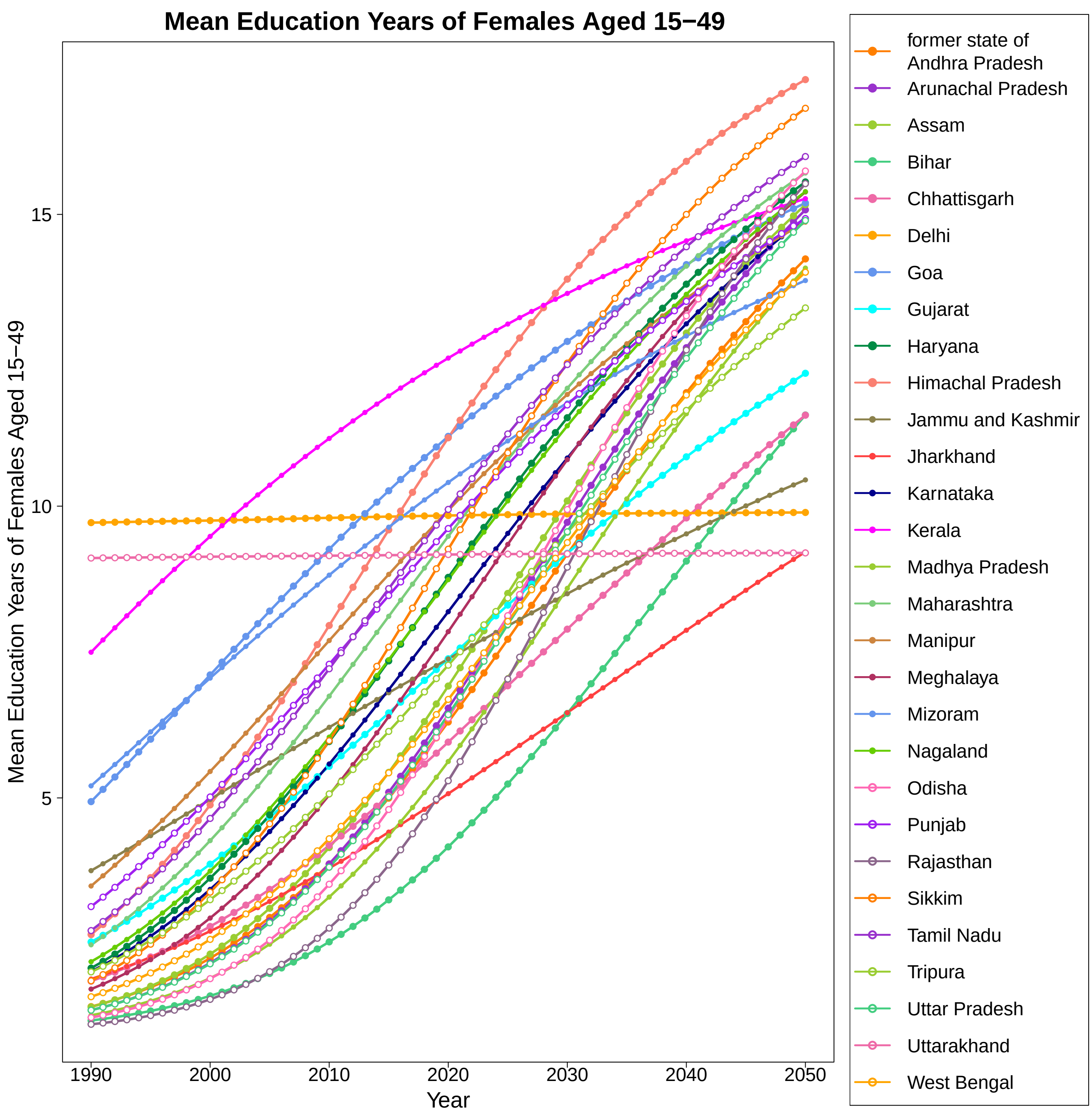


Figure 11: **Mean education years of females aged 15–49 by Indian state/UT model results.** Model medians are shown. States/UTs are differentiated by colors.

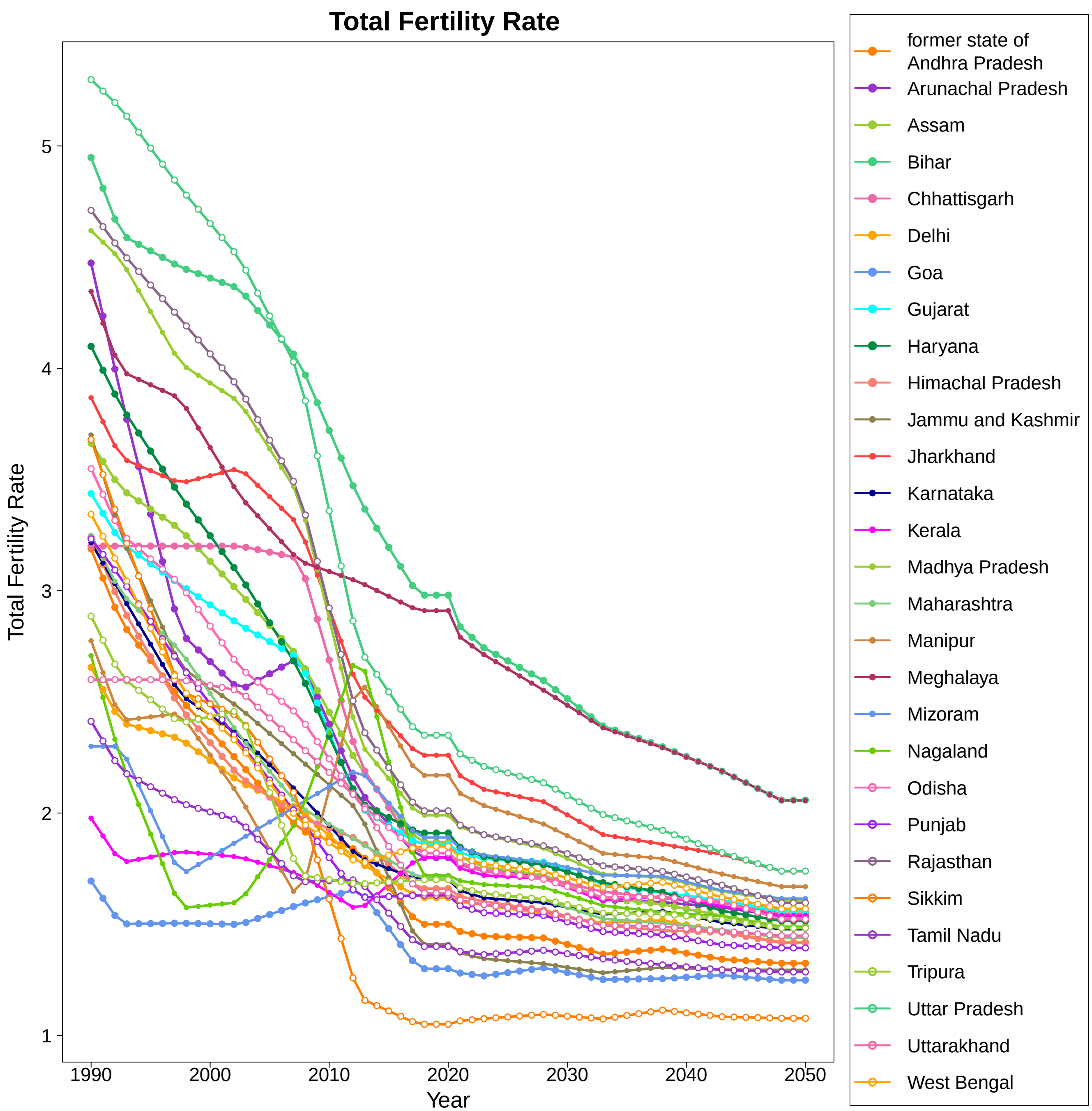
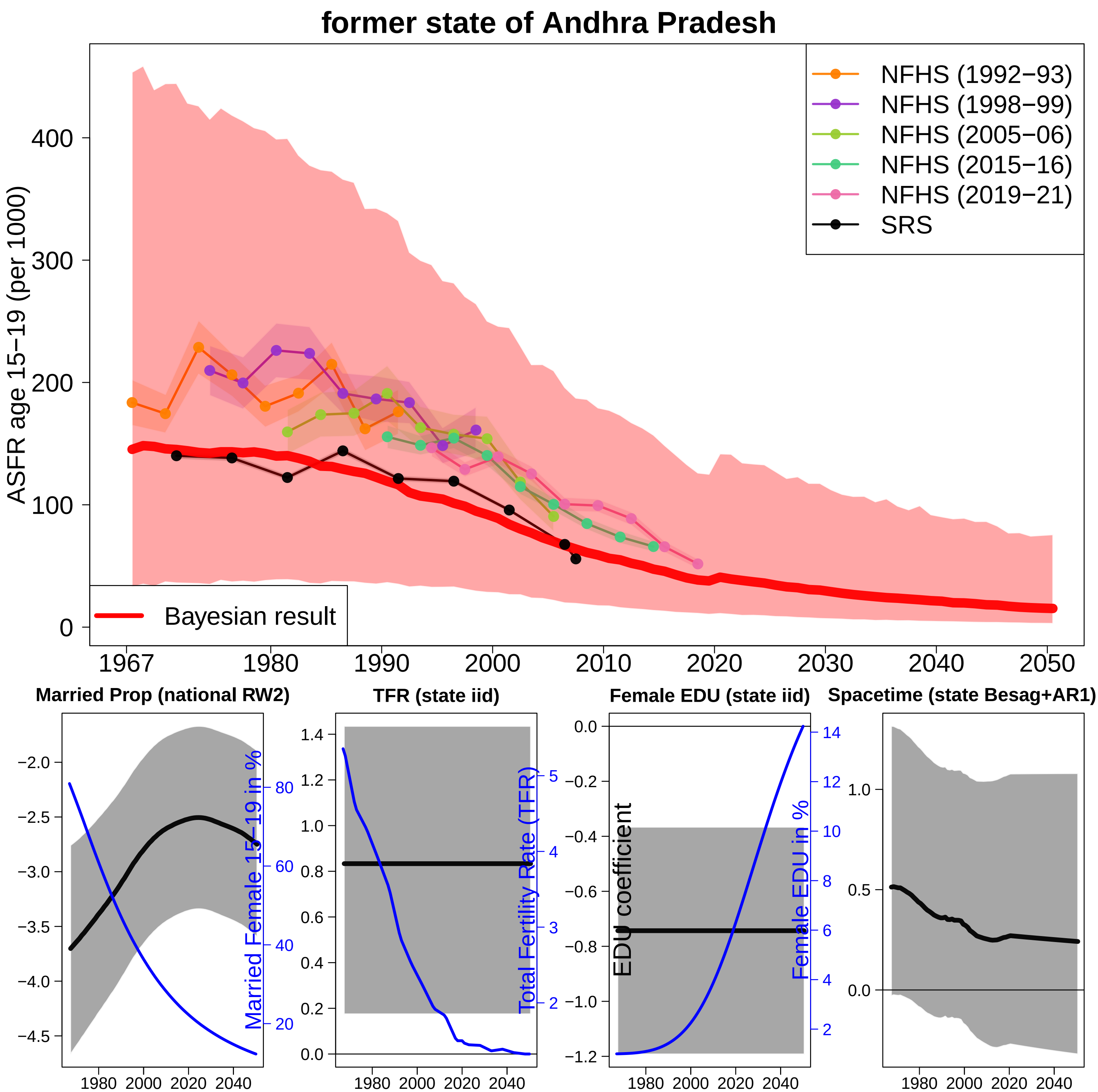
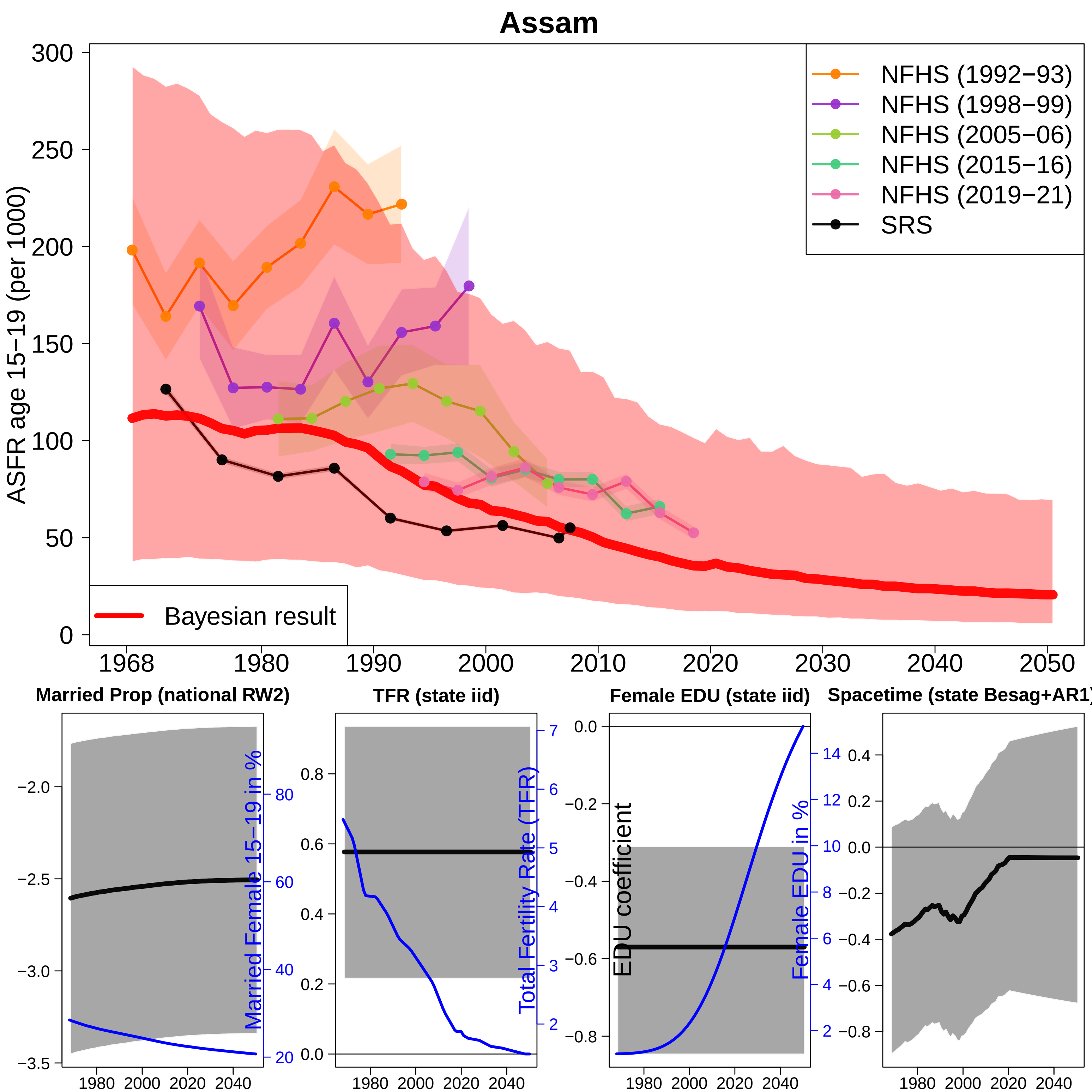


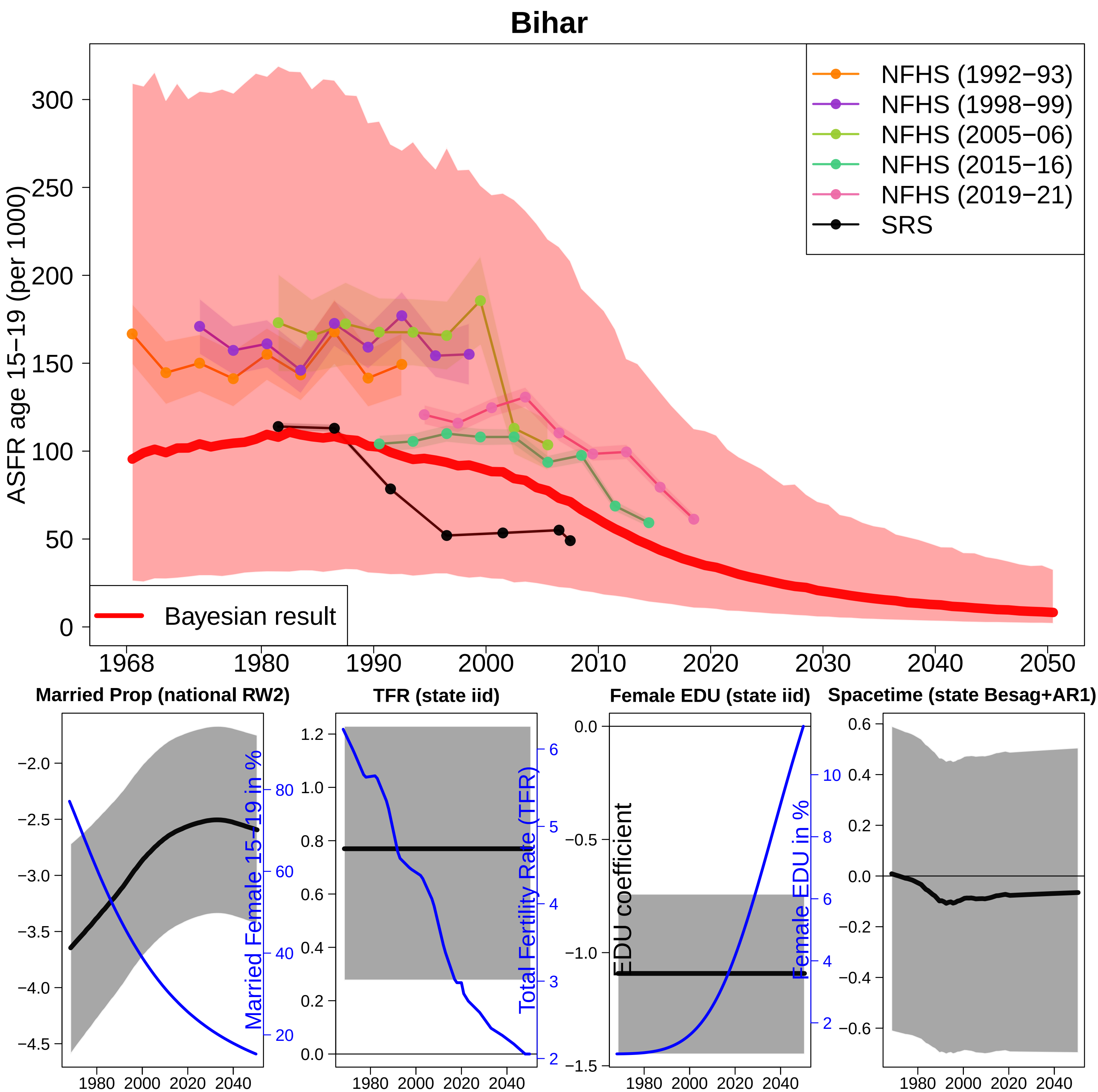
Figure 12: **Total fertility rate by Indian state/UT model results.** Model medians are shown. States/UTs are differentiated by colors.

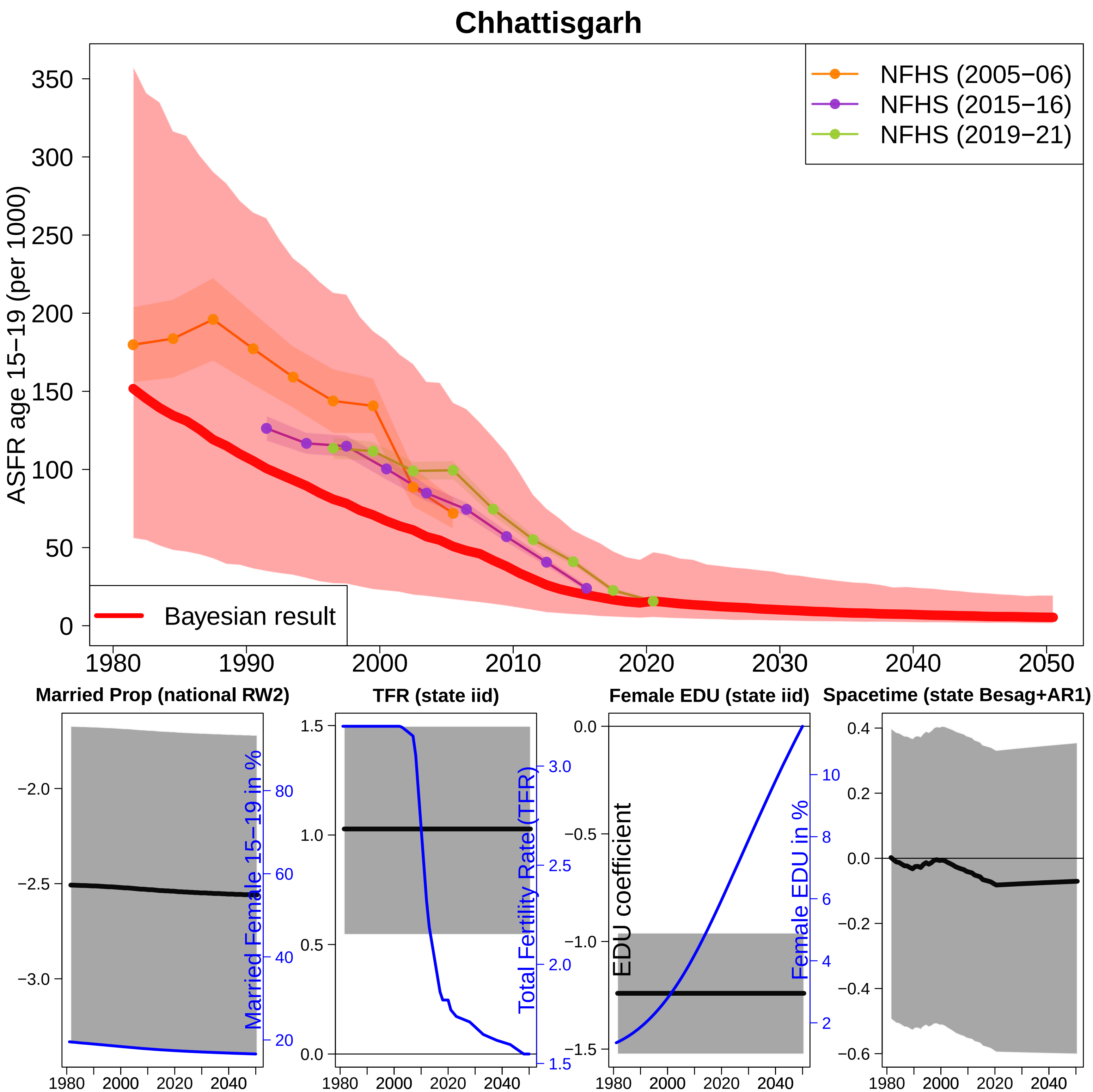
Figure 13: **ASFR age 15–19 estimates and projections by Indian state/UT, 1990–2050.** Red line and shades are the median and 95% credible intervals of the state-specific ASFR 15–19. ASFR 15–19 observations are displayed with dots and observations are connected with lines when obtained from the same source. Shades/vertical line segments around the data series represent the sampling variability in the series (quantified by two times the stochastic/sampling standard errors). Bayesian model parameter results and corresponding input covariates are shown at the bottom of each page.



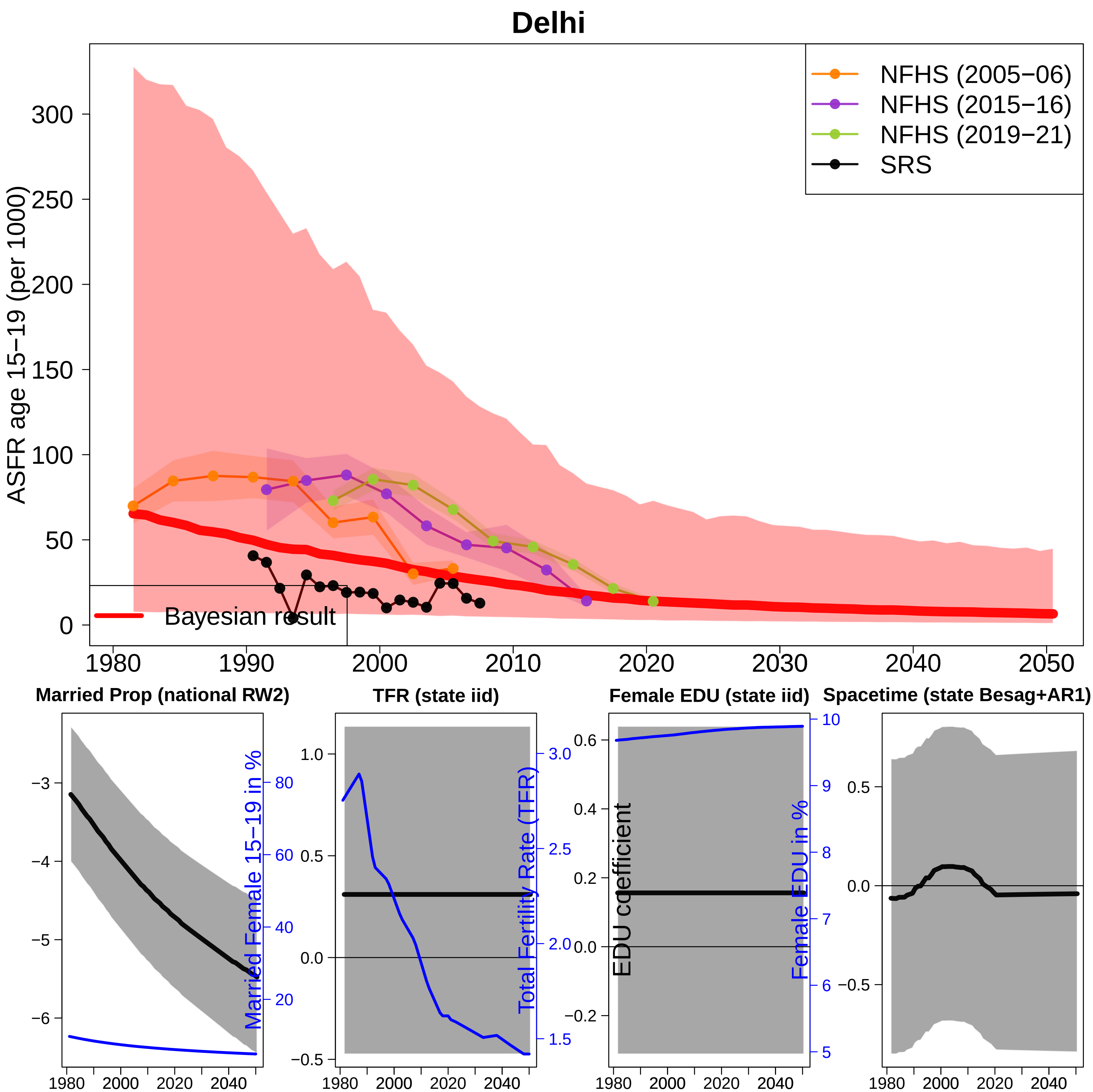
former state of Andhra Pradesh
ASFR age 15−19 (per 1000)
NFHS (1992−93)
NFHS (1998−99)
NFHS (2005−06)
NFHS (2015−16)
NFHS (2019−21)
SRS
Bayesian result
0
100
200
300
400
1967
1980
1990
2000
2010
2020
2030
2040
2050
Married Prop (national RW2)
Married Female 15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)


Assam
ASFR age 15−19 (per 1000)
NFHS (1992−93)
NFHS (1998−99)
NFHS (2005−06)
NFHS (2015−16)
NFHS (2019−21)
SRS
Bayesian result
Married Prop (national RW2)
Married Female 15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)

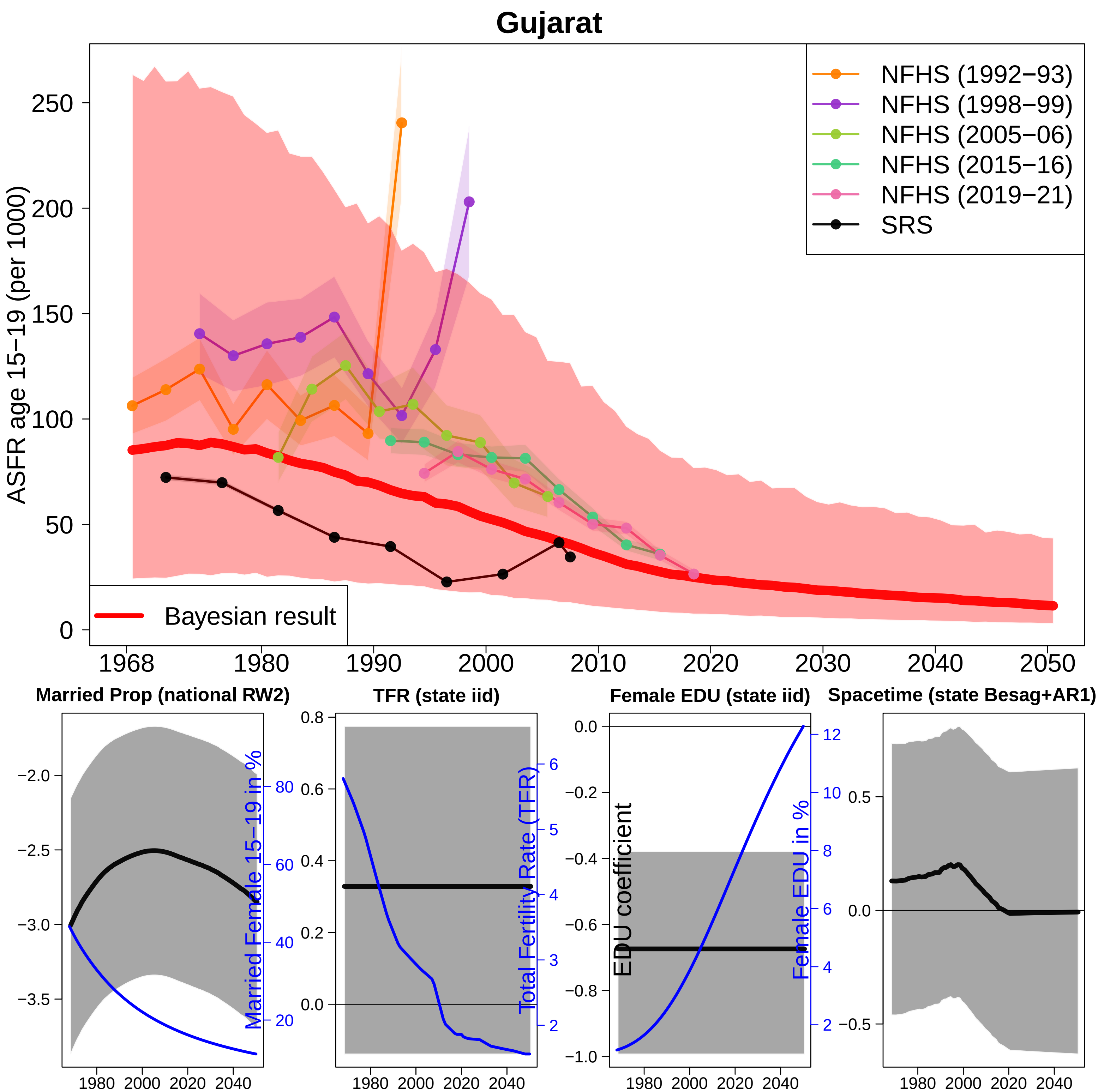

Bihar
ASFR age 15−19 (per 1000)
NFHS (1992−93)
NFHS (1998−99)
NFHS (2005−06)
NFHS (2015−16)
NFHS (2019−21)
SRS
Bayesian result
Married Prop (national RW2)
Married Female 15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)



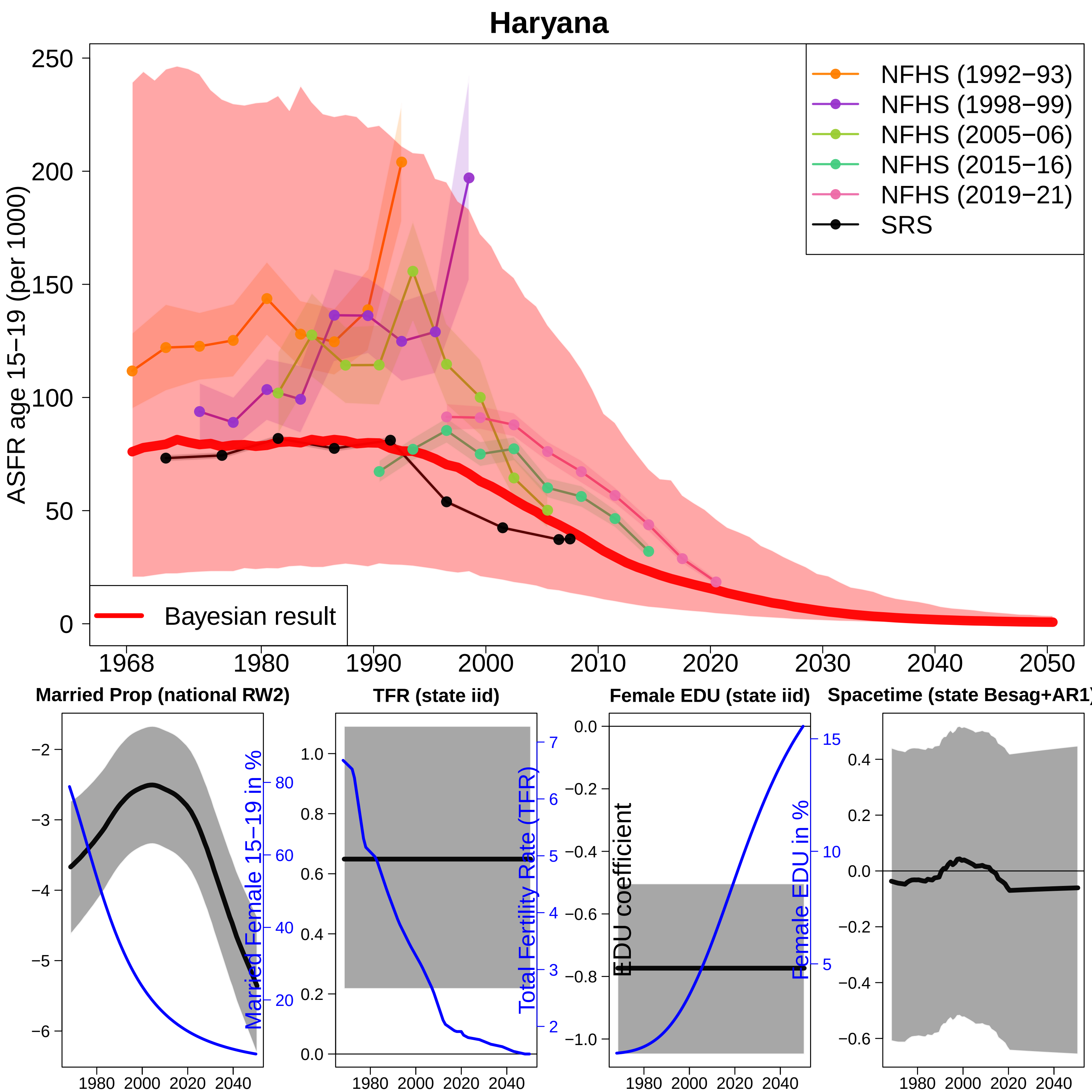
Chhattisgarh
NFHS (2005−06)
NFHS (2015−16)
NFHS (2019−21)
Bayesian result
ASFR age 15−19 (per 1000)
Married Prop (national RW2)
Married Female 15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)

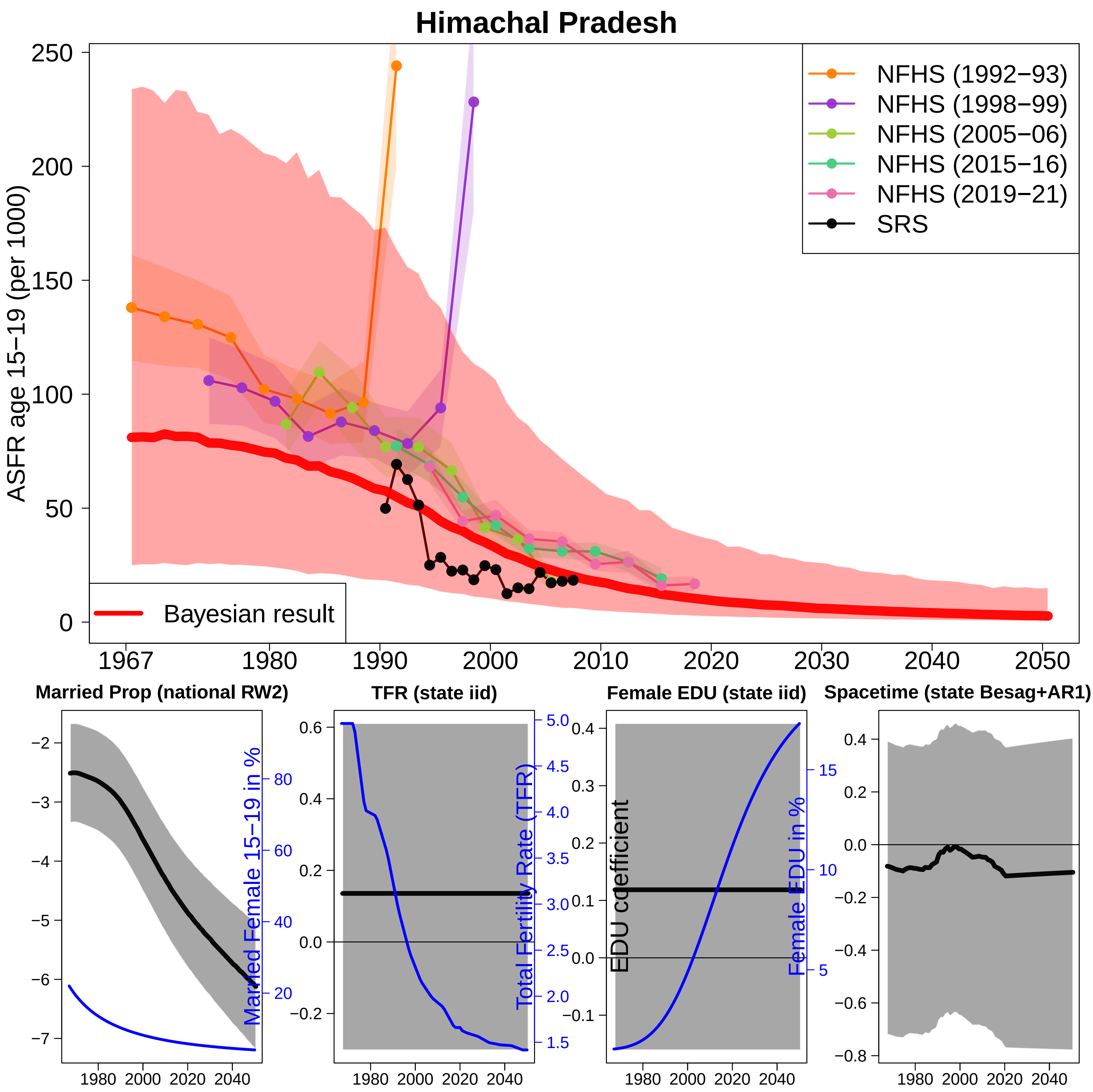


Delhi
NFHS (2005−06)
NFHS (2015−16)
NFHS (2019−21)
SRS
Bayesian result
ASFR age 15−19 (per 1000)
Married Prop (national RW2)
Married Female 15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)

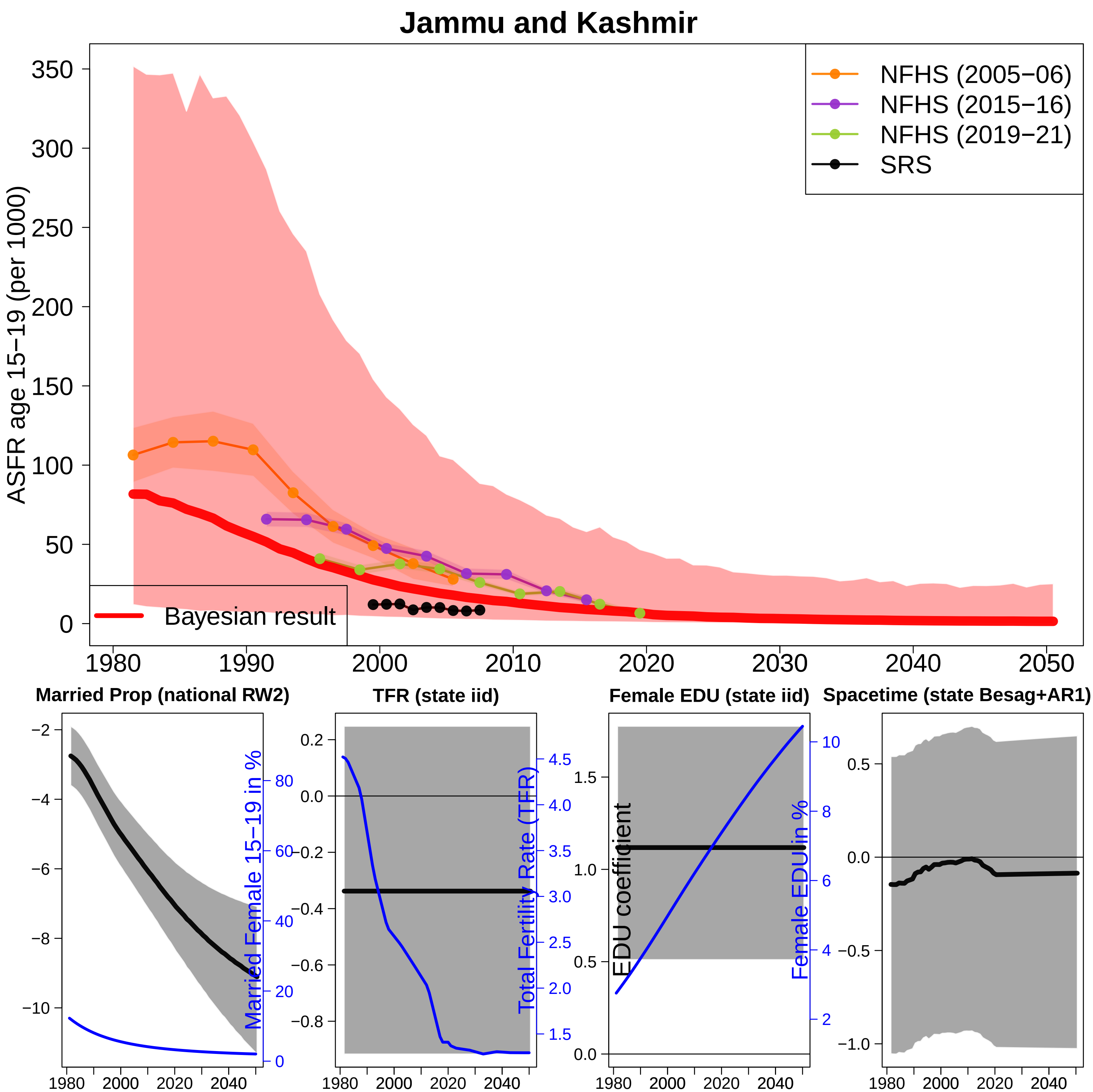


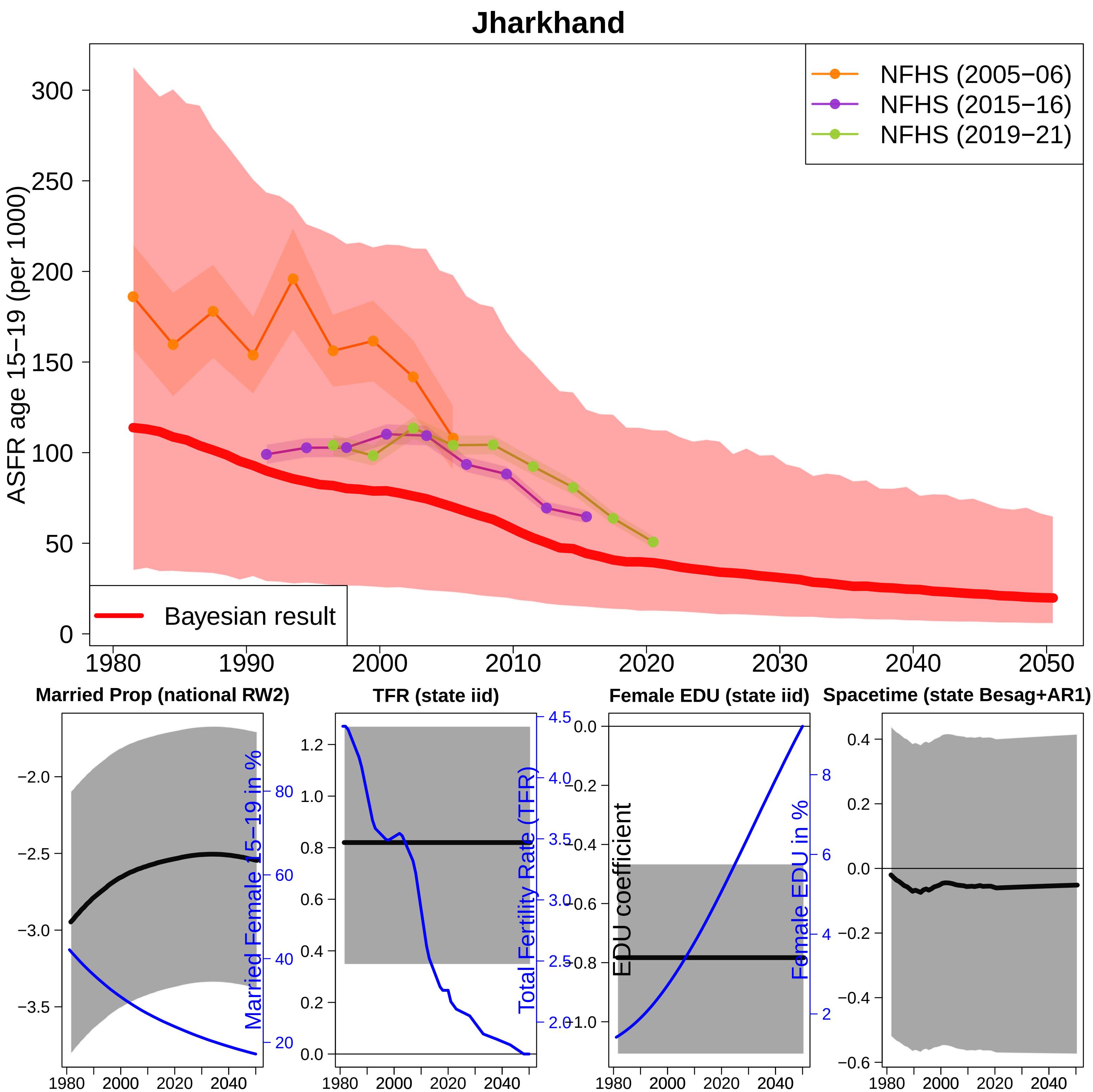
Gujarat
ASFR age 15−19 (per 1000)
NFHS (1992−93)
NFHS (1998−99)
NFHS (2005−06)
NFHS (2015−16)
NFHS (2019−21)
SRS
Bayesian result
Married Prop (national RW2)
Married Female 15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)

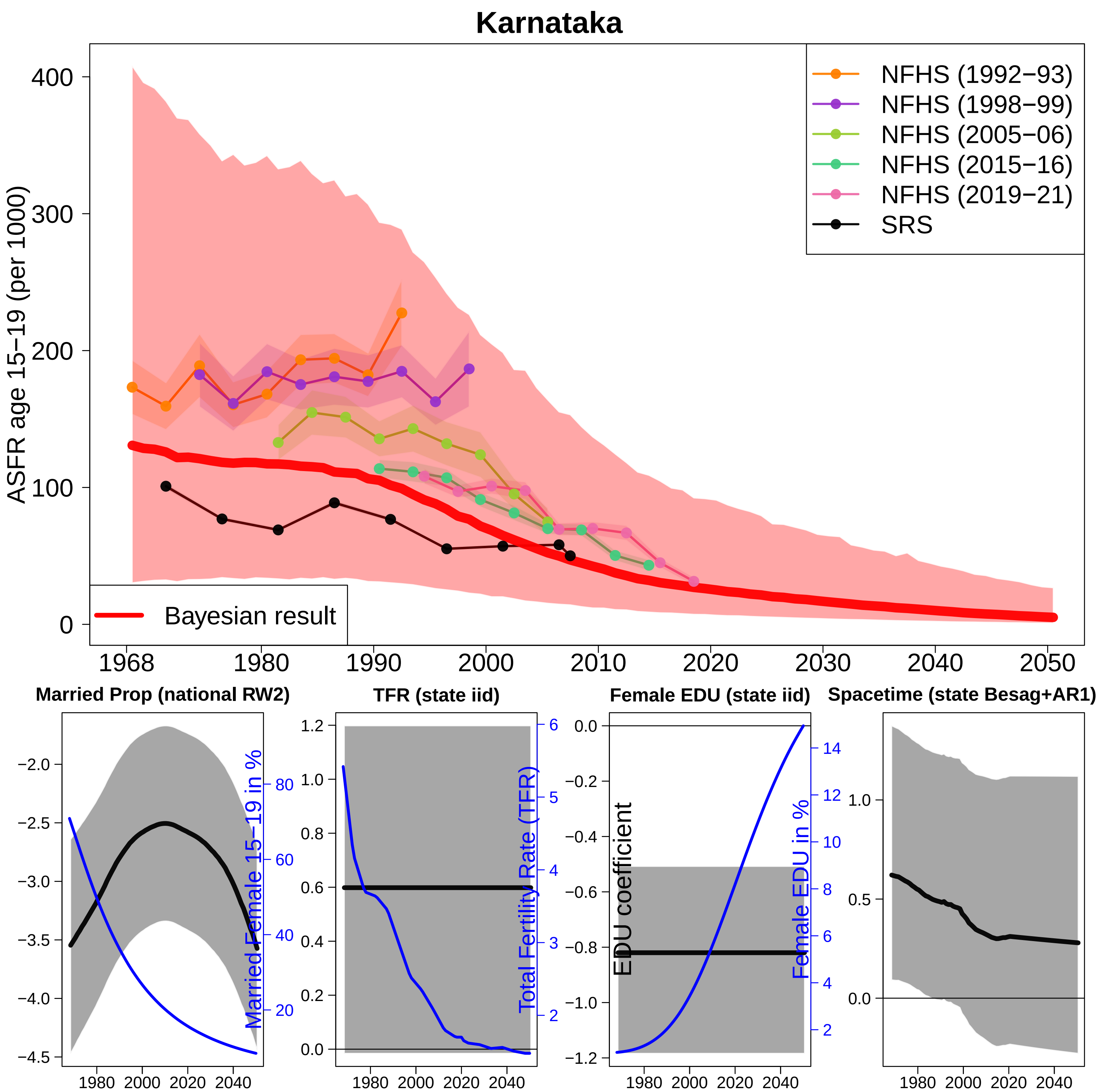

Haryana
ASFR age 15−19 (per 1000)
NFHS (1992−93)
NFHS (1998−99)
NFHS (2005−06)
NFHS (2015−16)
NFHS (2019−21)
SRS
Bayesian result
Married Prop (national RW2)
Married Female 15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)

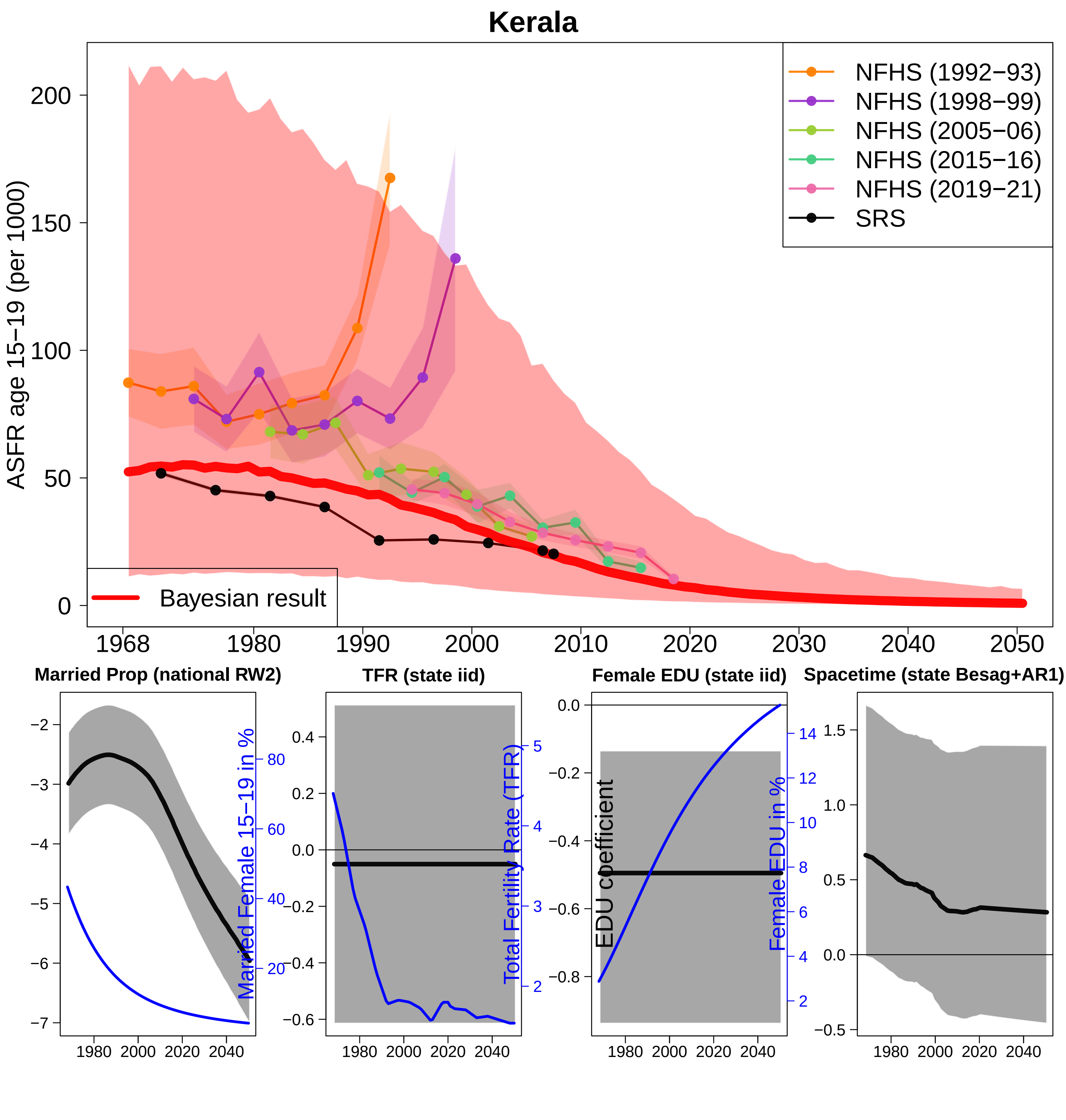


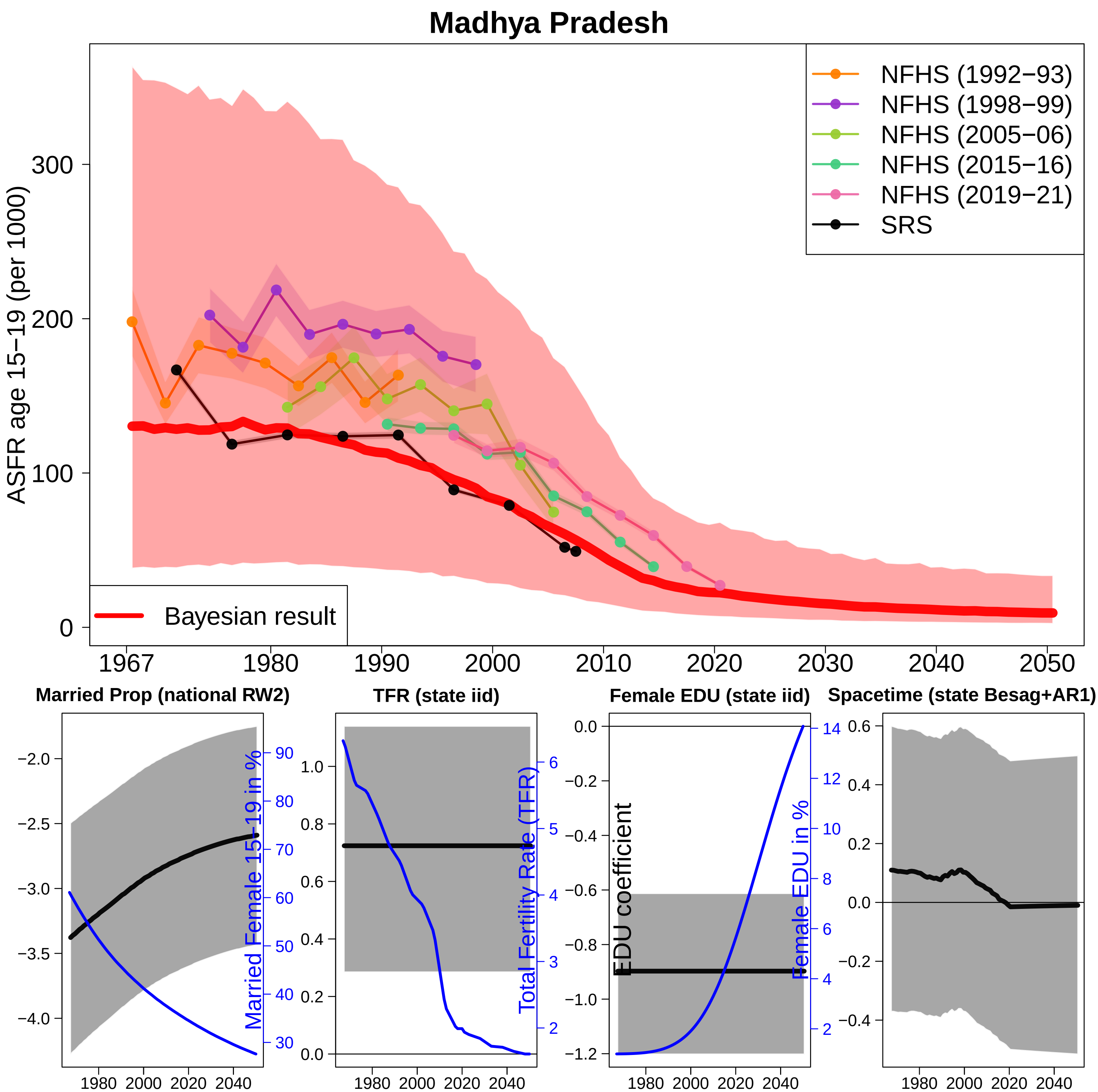
Himachal Pradesh
ASFR age 15−19 (per 1000)
NFHS (1992−93)
NFHS (1998−99)
NFHS (2005−06)
NFHS (2015−16)
NFHS (2019−21)
SRS
Bayesian result
Married Prop (national RW2)
Married Female 15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)


Jammu and Kashmir
ASFR age 15−19 (per 1000)
NFHS (2005−06)
NFHS (2015−16)
NFHS (2019−21)
SRS
Bayesian result
Married Prop (national RW2)
Married Female 15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)

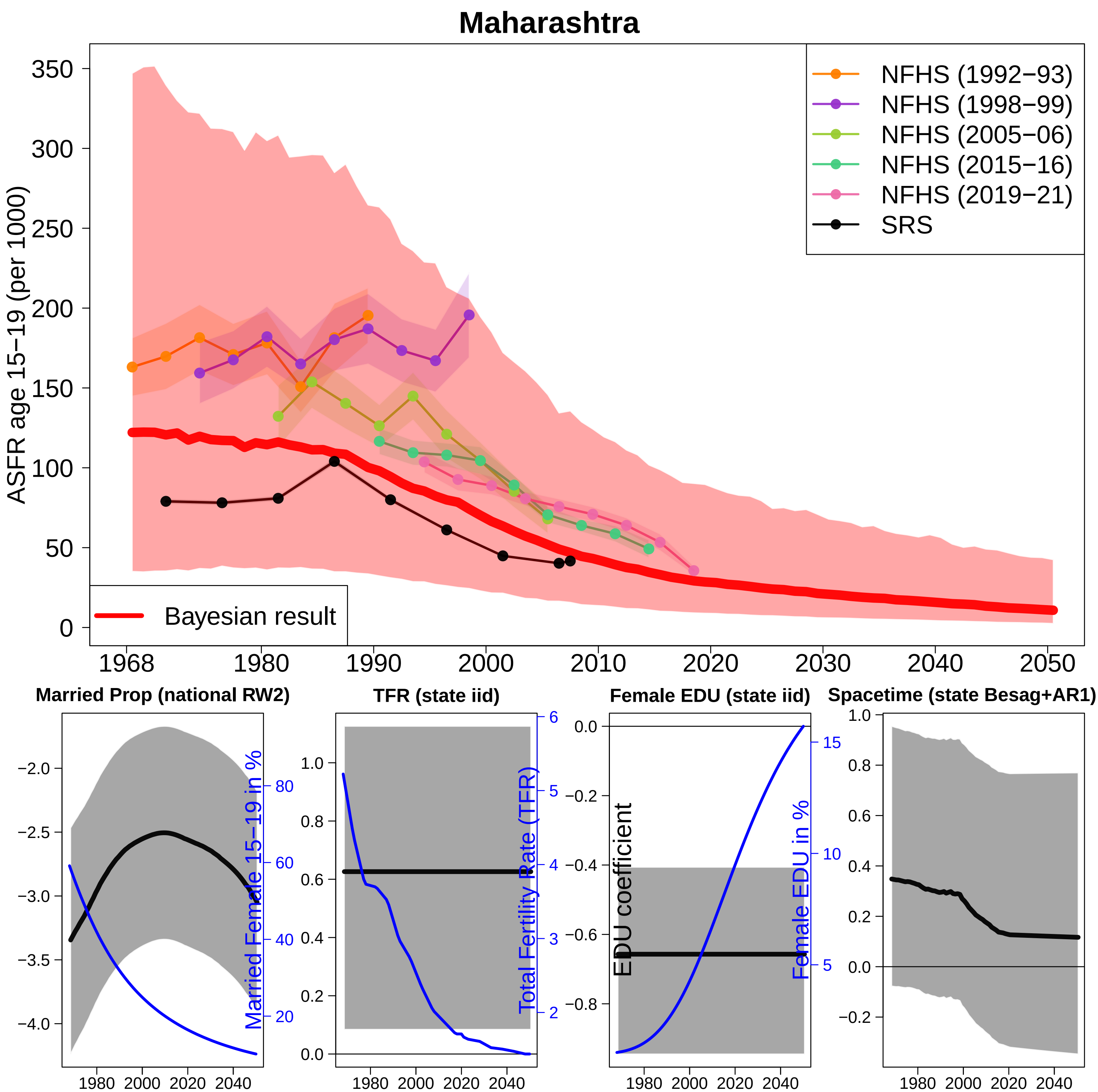

Jharkhand
NFHS (2005−06)
NFHS (2015−16)
NFHS (2019−21)
Bayesian result
ASFR age 15−19 (per 1000)
Married Prop (national RW2)
Married Female 15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)

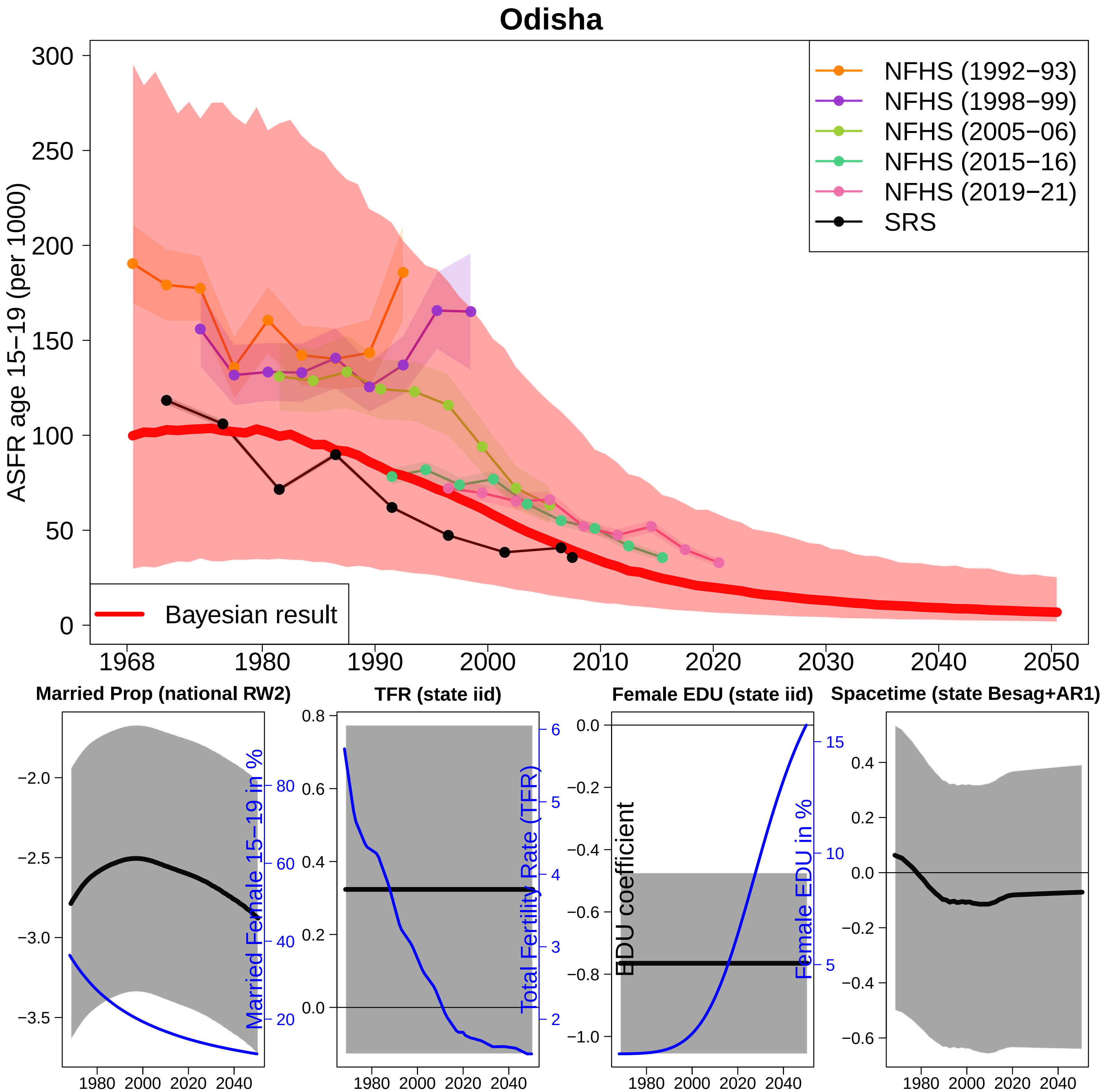

Karnataka
ASFR age 15−19 (per 1000)
NFHS (1992−93)
NFHS (1998−99)
NFHS (2005−06)
NFHS (2015−16)
NFHS (2019−21)
SRS
Bayesian result
Married Prop (national RW2)
Married Female 15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)

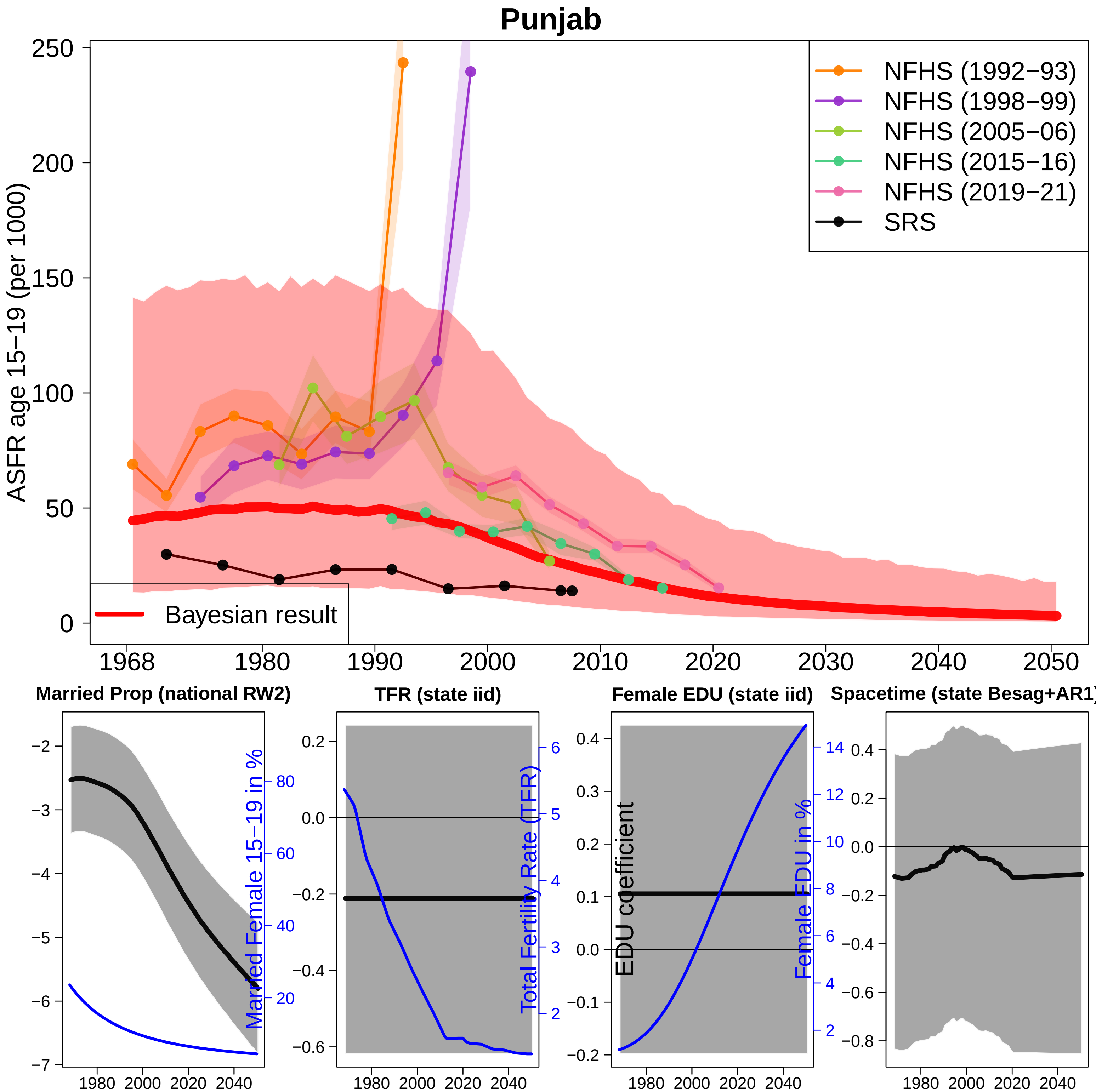

Kerala
ASFR age 15−19 (per 1000)
NFHS (1992−93)
NFHS (1998−99)
NFHS (2005−06)
NFHS (2015−16)
NFHS (2019−21)
SRS
Bayesian result
Married Prop (national RW2)
Married Female 15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)



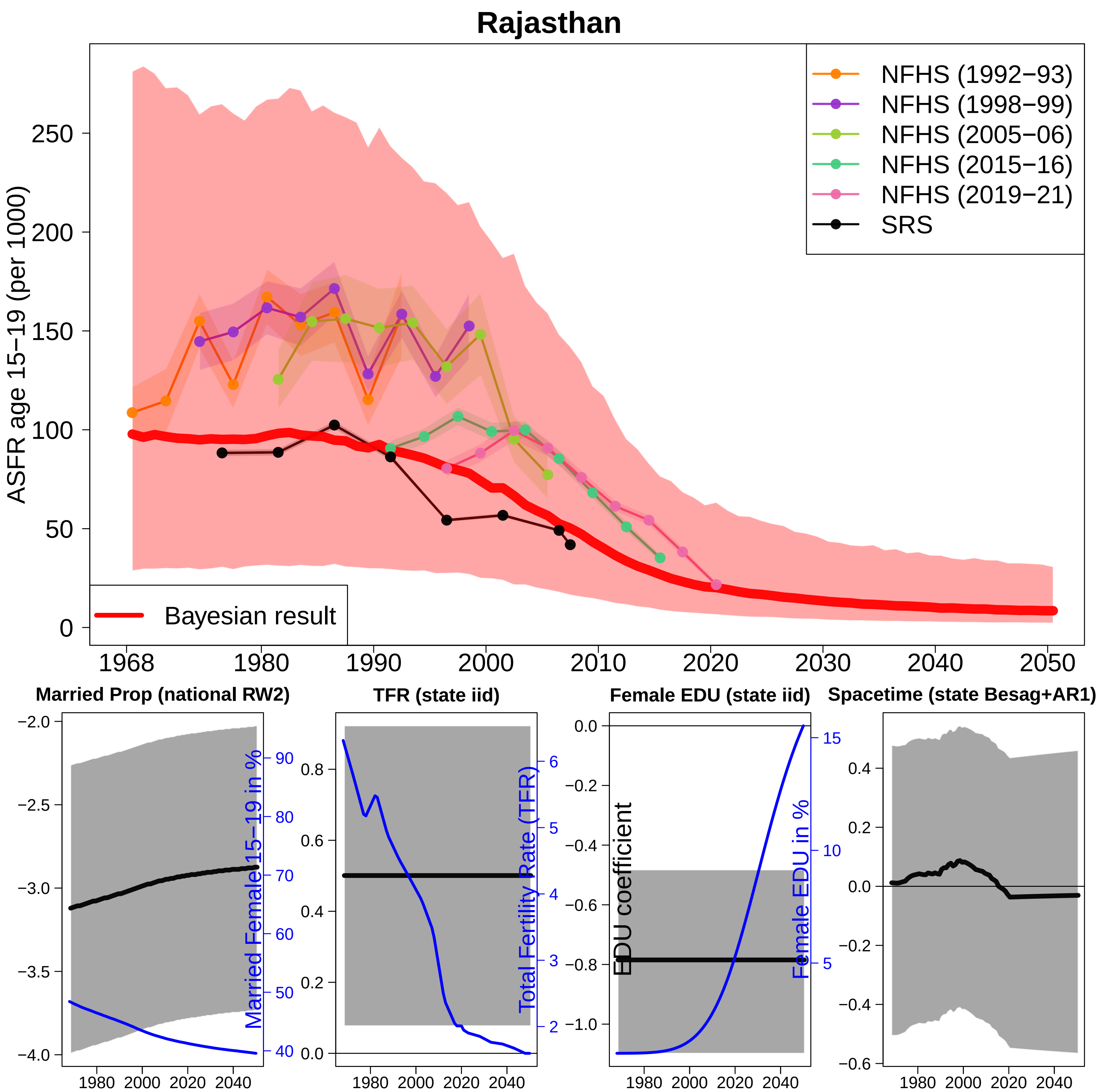
Madhya Pradesh
NFHS (1992−93)
NFHS (1998−99)
NFHS (2005−06)
NFHS (2015−16)
NFHS (2019−21)
SRS
Bayesian result
ASFR age 15−19 (per 1000)
0
100
200
300
1967
1980
1990
2000
2010
2020
2030
2040
2050
Married Prop (national RW2)
Married Female 15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)
1980 2000 2020 2040

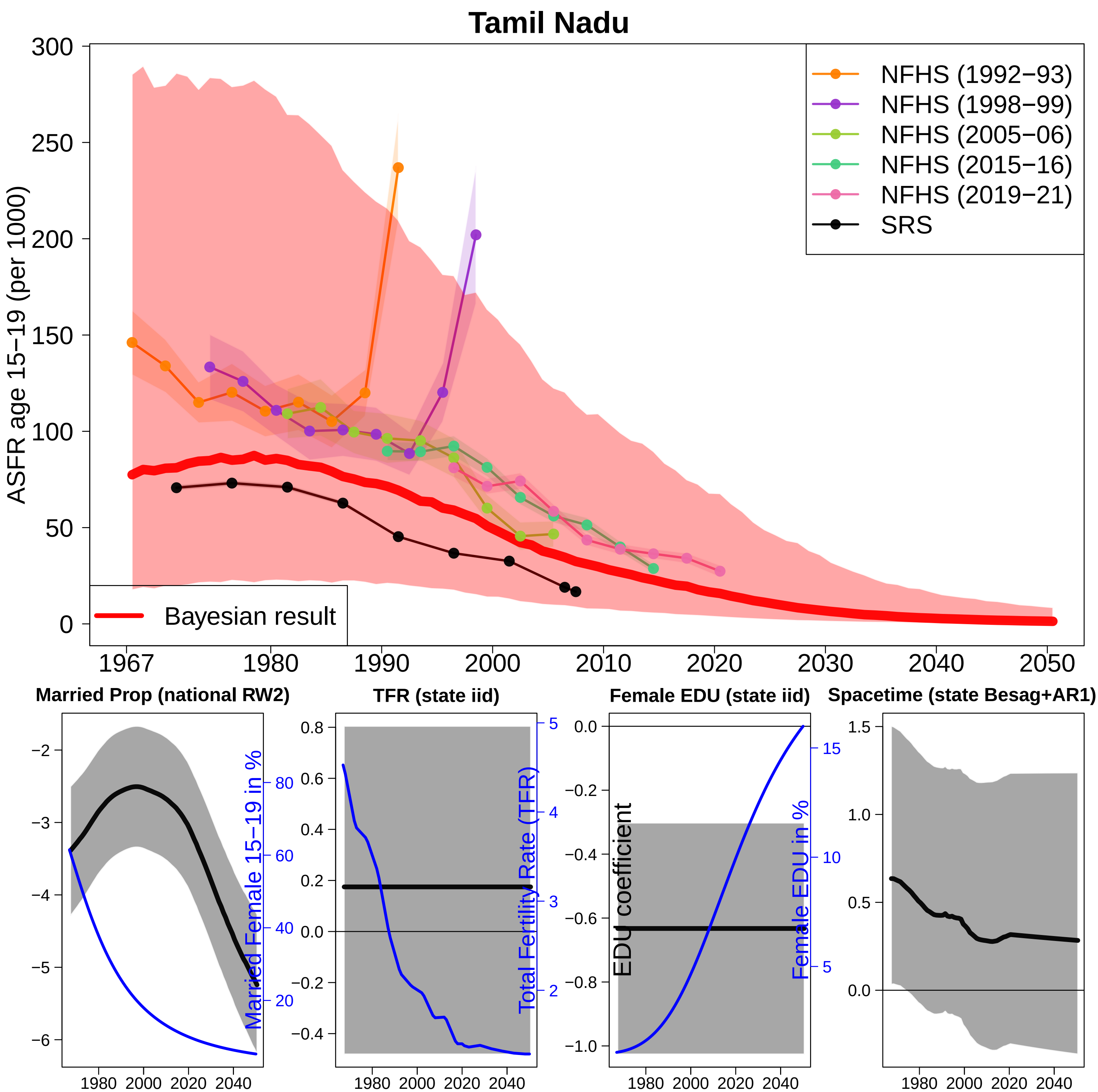

Maharashtra
ASFR age 15−19 (per 1000)
NFHS (1992−93)
NFHS (1998−99)
NFHS (2005−06)
NFHS (2015−16)
NFHS (2019−21)
SRS
Bayesian result
Married Prop (national RW2)
Married Female 15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)


Odisha
ASFR age 15−19 (per 1000)
NFHS (1992−93)
NFHS (1998−99)
NFHS (2005−06)
NFHS (2015−16)
NFHS (2019−21)
SRS
Bayesian result
Married Prop (national RW2)
Married Female 15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)

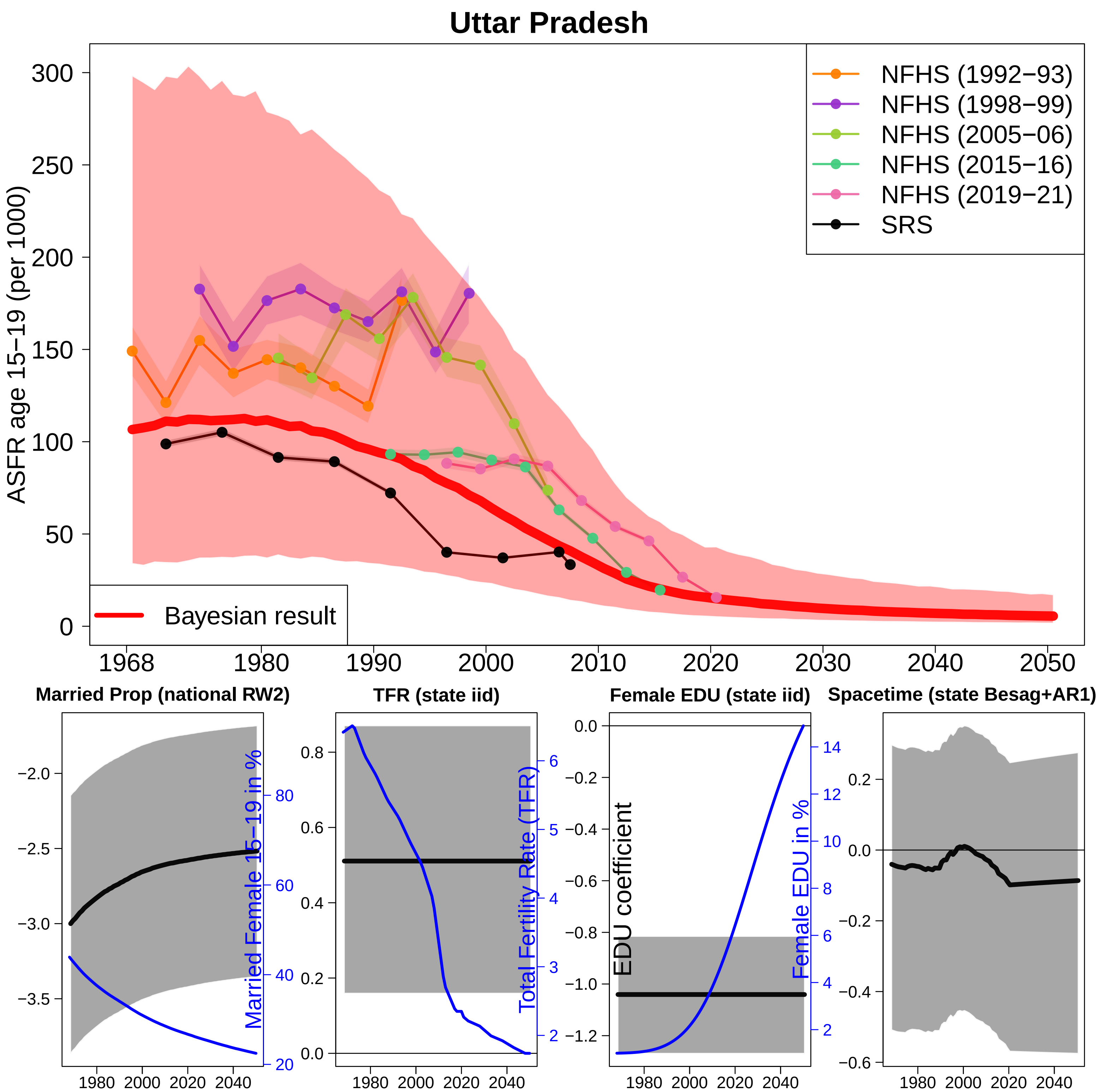


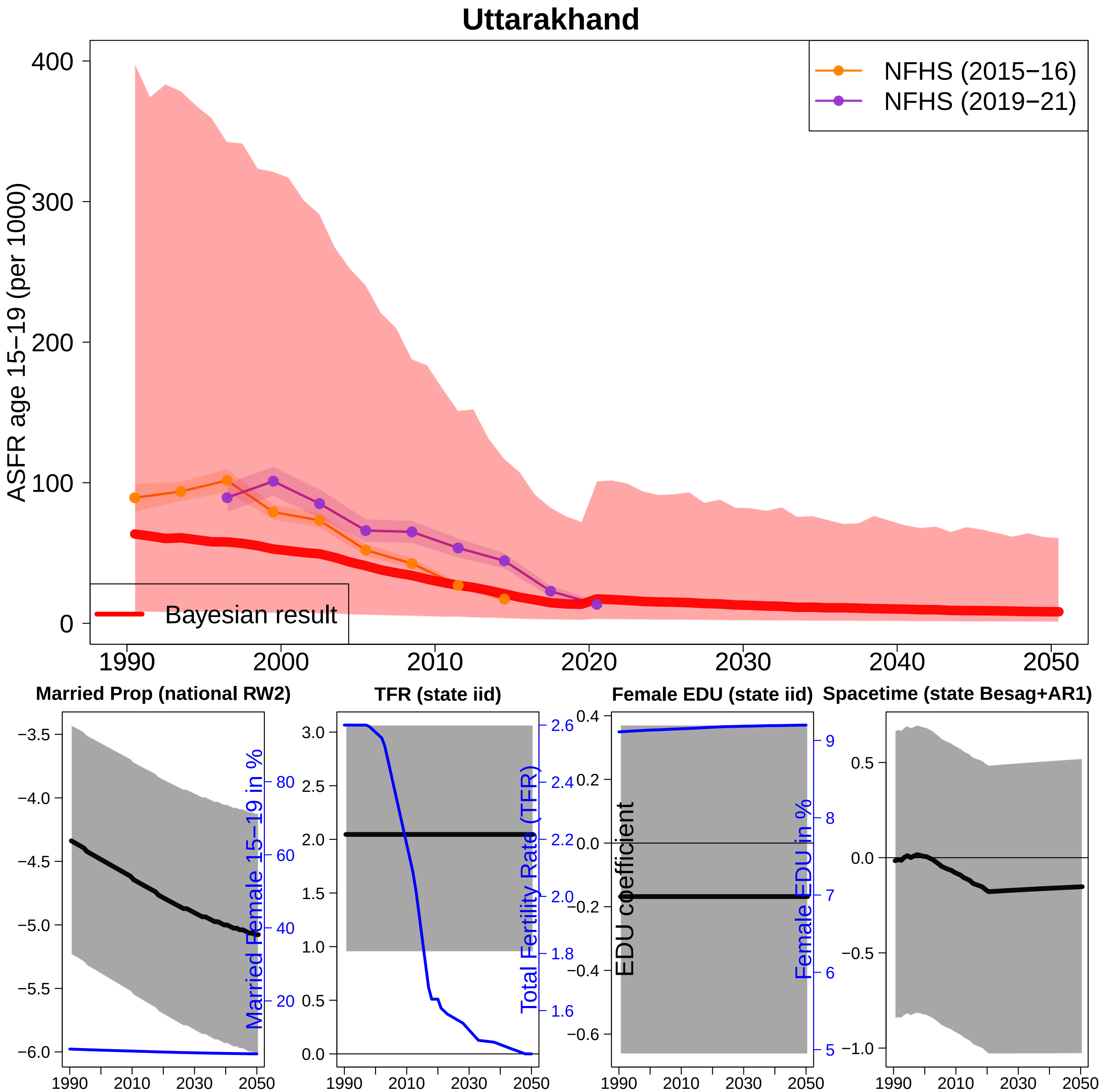
Punjab
ASFR age 15−19 (per 1000)
NFHS (1992−93)
NFHS (1998−99)
NFHS (2005−06)
NFHS (2015−16)
NFHS (2019−21)
SRS
Bayesian result
Married Prop (national RW2)
Married Female 15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)

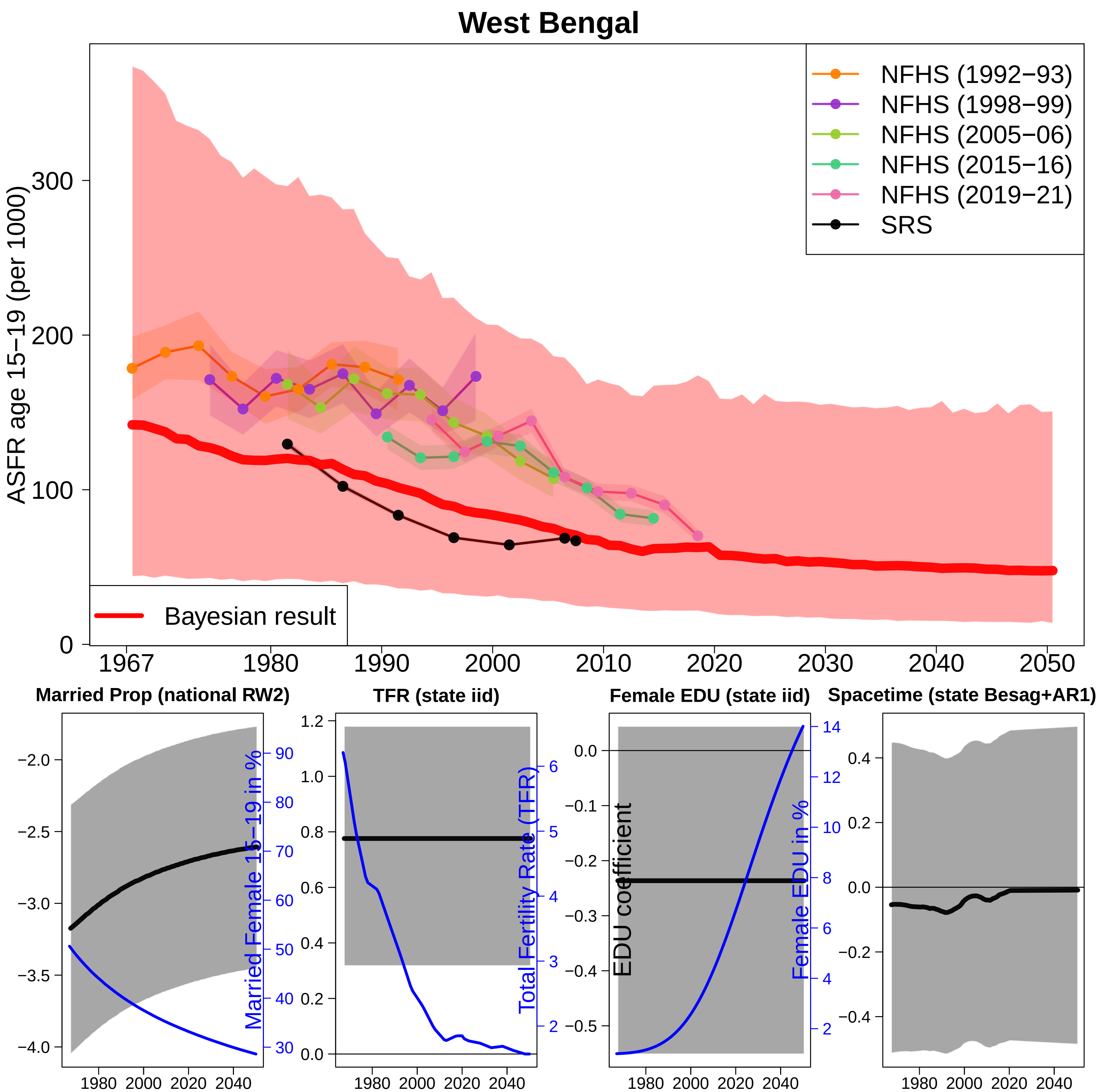

Rajasthan
NFHS (1992−93)
NFHS (1998−99)
NFHS (2005−06)
NFHS (2015−16)
NFHS (2019−21)
SRS
Bayesian result
ASFR age 15−19 (per 1000)
Married Prop (national RW2)
Married Female15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)



Tamil Nadu
ASFR age 15−19 (per 1000)
NFHS (1992−93)
NFHS (1998−99)
NFHS (2005−06)
NFHS (2015−16)
NFHS (2019−21)
SRS
Bayesian result
Married Prop (national RW2)
Married Female 15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)


Uttar Pradesh
ASFR age 15−19 (per 1000)
NFHS (1992−93)
NFHS (1998−99)
NFHS (2005−06)
NFHS (2015−16)
NFHS (2019−21)
SRS
Bayesian result
Married Prop (national RW2)
Married Female 15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)


Uttarakhand
NFHS (2015−16)
NFHS (2019−21)
Bayesian result
ASFR age 15−19 (per 1000)
Married Prop (national RW2)
Married Female 15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)



West Bengal
ASFR age 15−19 (per 1000)
NFHS (1992−93)
NFHS (1998−99)
NFHS (2005−06)
NFHS (2015−16)
NFHS (2019−21)
SRS
Bayesian result
Married Prop (national RW2)
Married Female 15−19 in %
TFR (state iid)
Total Fertility Rate (TFR)
Female EDU (state iid)
EDU coefficient
Female EDU in %
Spacetime (state Besag+AR1)

— The End —